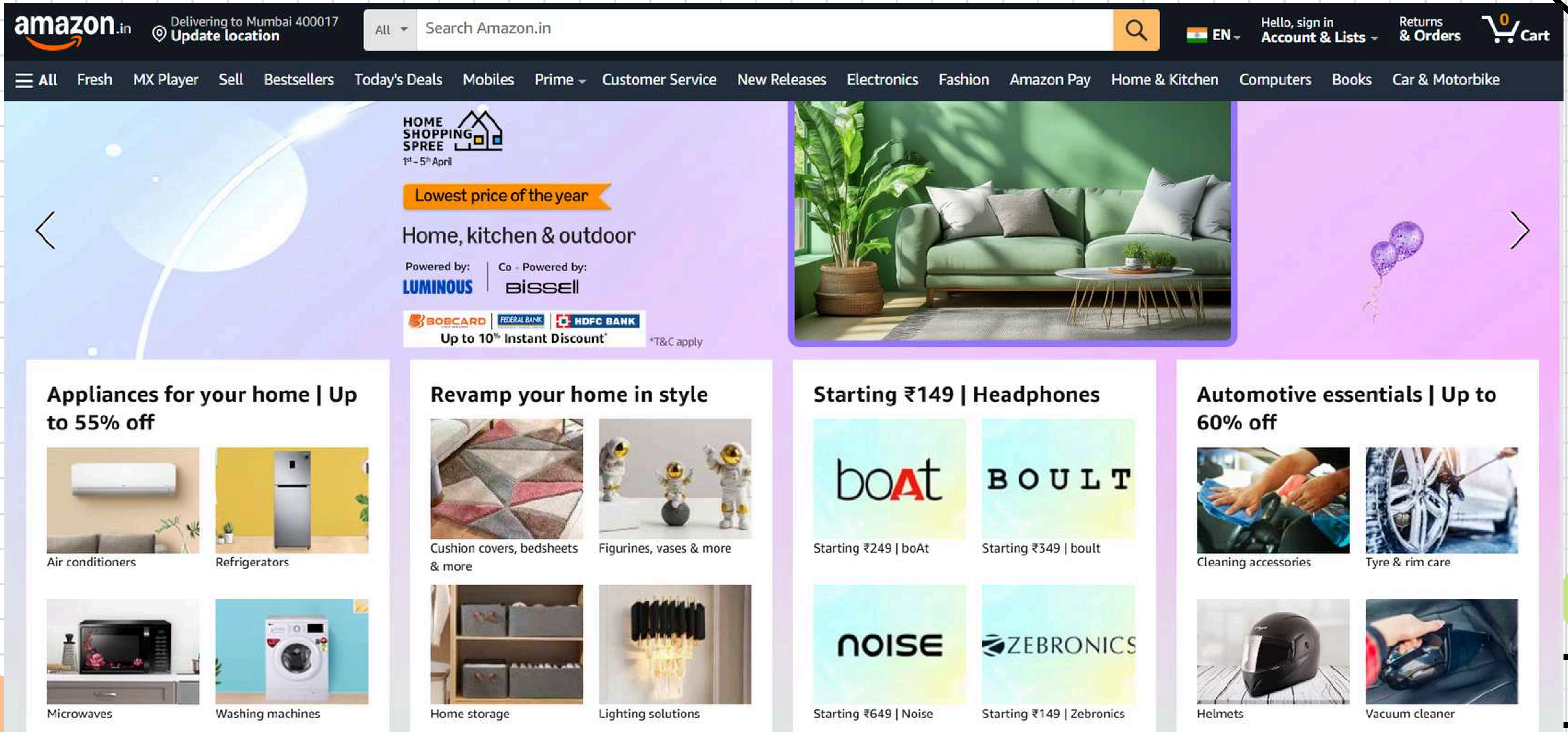
WHAT IS FRONT-END DEVELOPMENT?


amazon.in
Delivering to Mumbai 400017
Update location
All
Search Amazon.in
EN
Hello, sign in
Account & Lists
Returns
& Orders
0
Cart
All
Fresh
MX Player
Sell
Bestsellers
Today's Deals
Mobiles
Prime
Customer Service
New Releases
Electronics
Fashion
Amazon Pay
Home & Kitchen
Computers
Books
Car & Motorbike
HOME SHOPPING SPREE
1st - 5th April
Lowest price of the year
Home, kitchen & outdoor
Powered by:
LUMINOUS
Co - Powered by:
BISSELL
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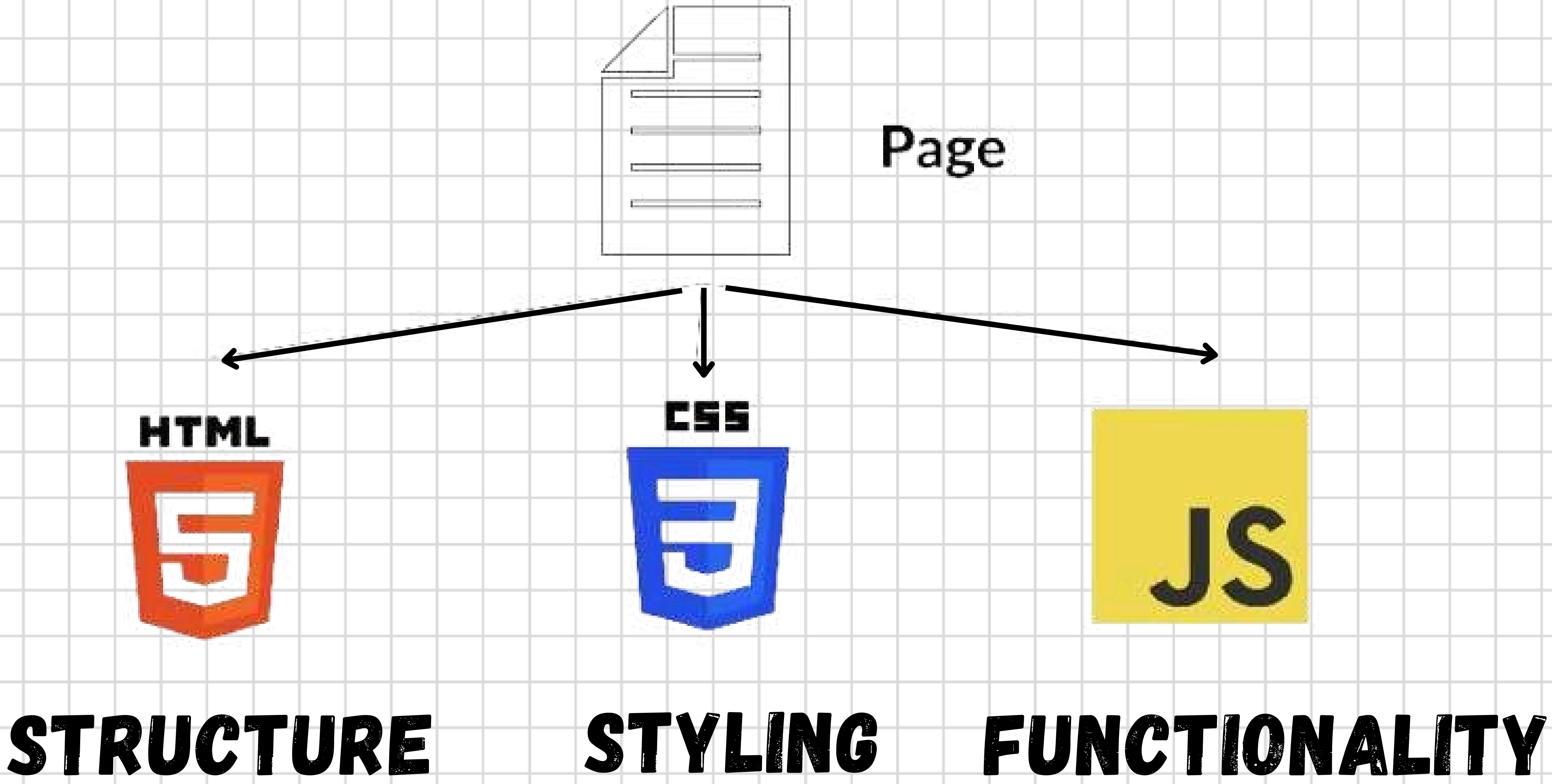
HDFC BANK
Up to 10% Instant Discount*
*T&C apply
Appliances for your home | Up to 55% off
Air conditioners
Refrigerators
Microwaves
Washing machines
Revamp your home in style
Cushion covers, bedsheets & more
Figurines, vases & more
Home storage
Lighting solutions
Starting ₹149 | Headphones
boAt
BOULT
Starting ₹249 | boAt
Starting ₹349 | boult
NOISE
ZEBRONICS
Starting ₹649 | Noise
Starting ₹149 | Zebronics
Automotive essentials | Up to 60% off
Cleaning accessories
Tyre & rim care
Helmets
Vacuum cleaner


Page
HTML
CSS
JS
STRUCTURE
STYLING
FUNCTIONALITY

TRADITIONAL VS MODERN FRAMEWORKS

Traditional

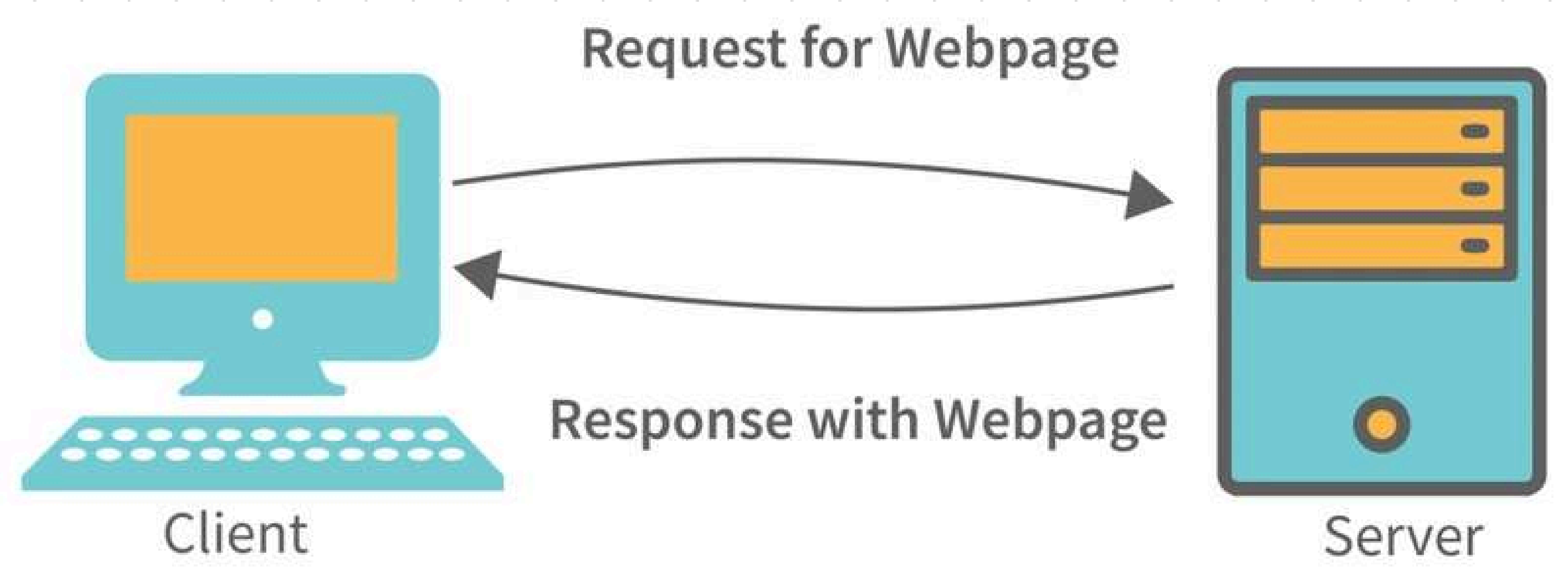
HTML JS CSS

Modern frameworks

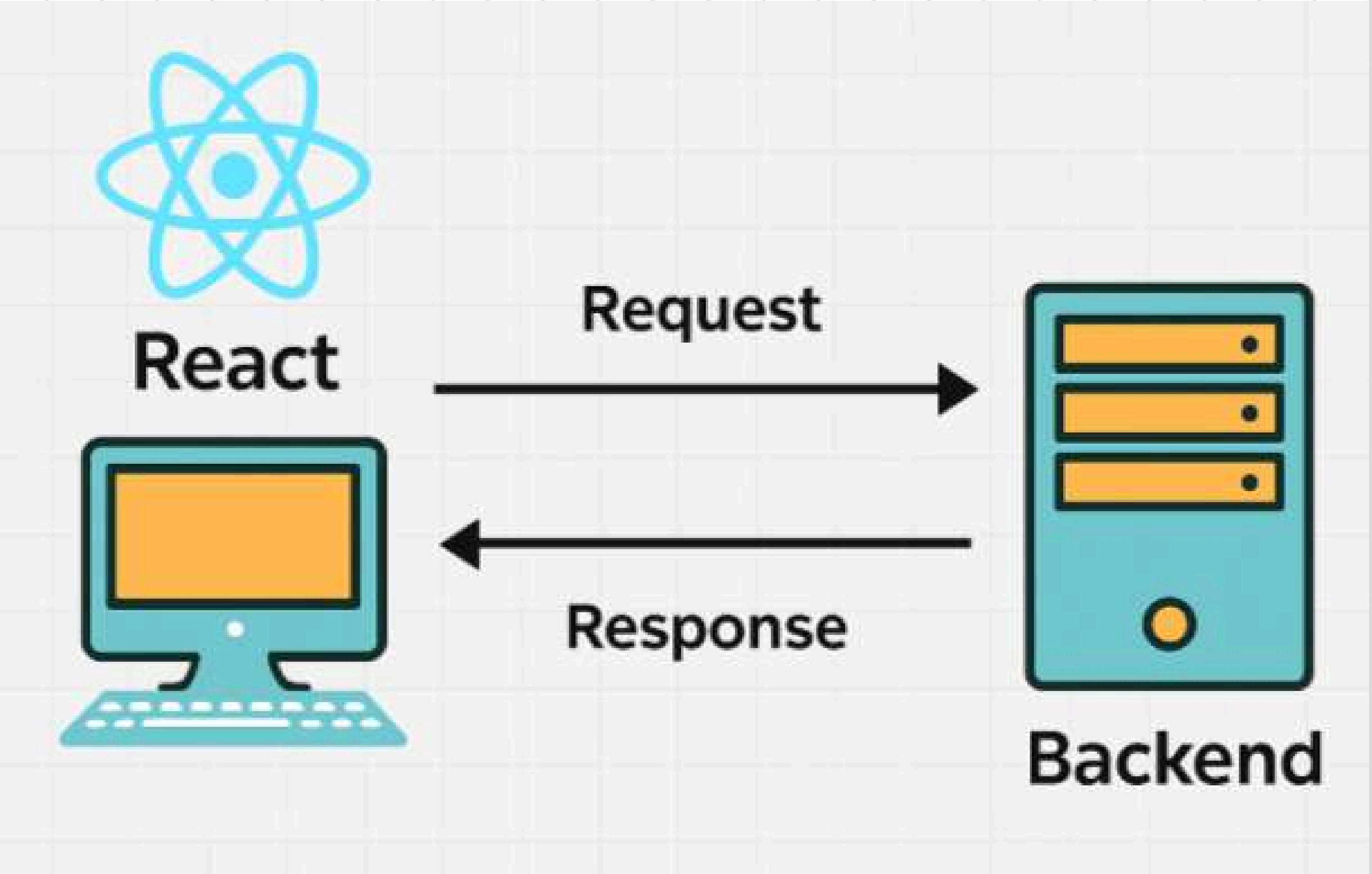
AngularJS ReactJS

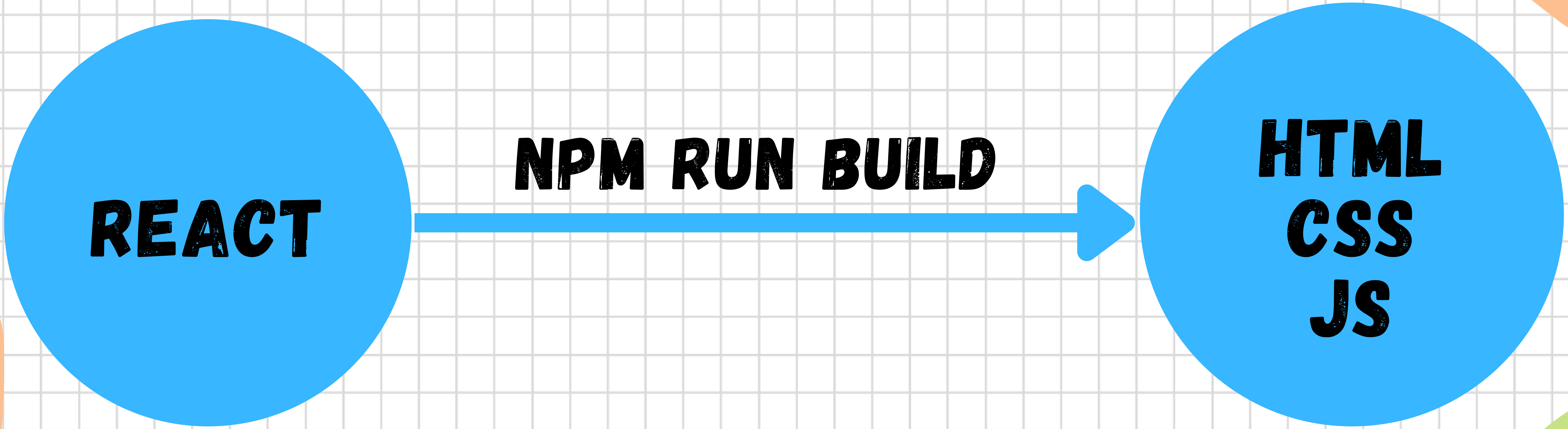
Vue.js

CLIENT-SERVER ARCHITECTURE



HOW REACT INTERACTS WITH THE BACKEND ?




REACT
NPM RUN BUILD
HTML
CSS
JS

WHAT IS REACT?

REACT IS..

JAVASCRIPT LIBRARY

DEVELOPED BY FACEBOOK

OPEN SOURCE

USED TO CREATE UI

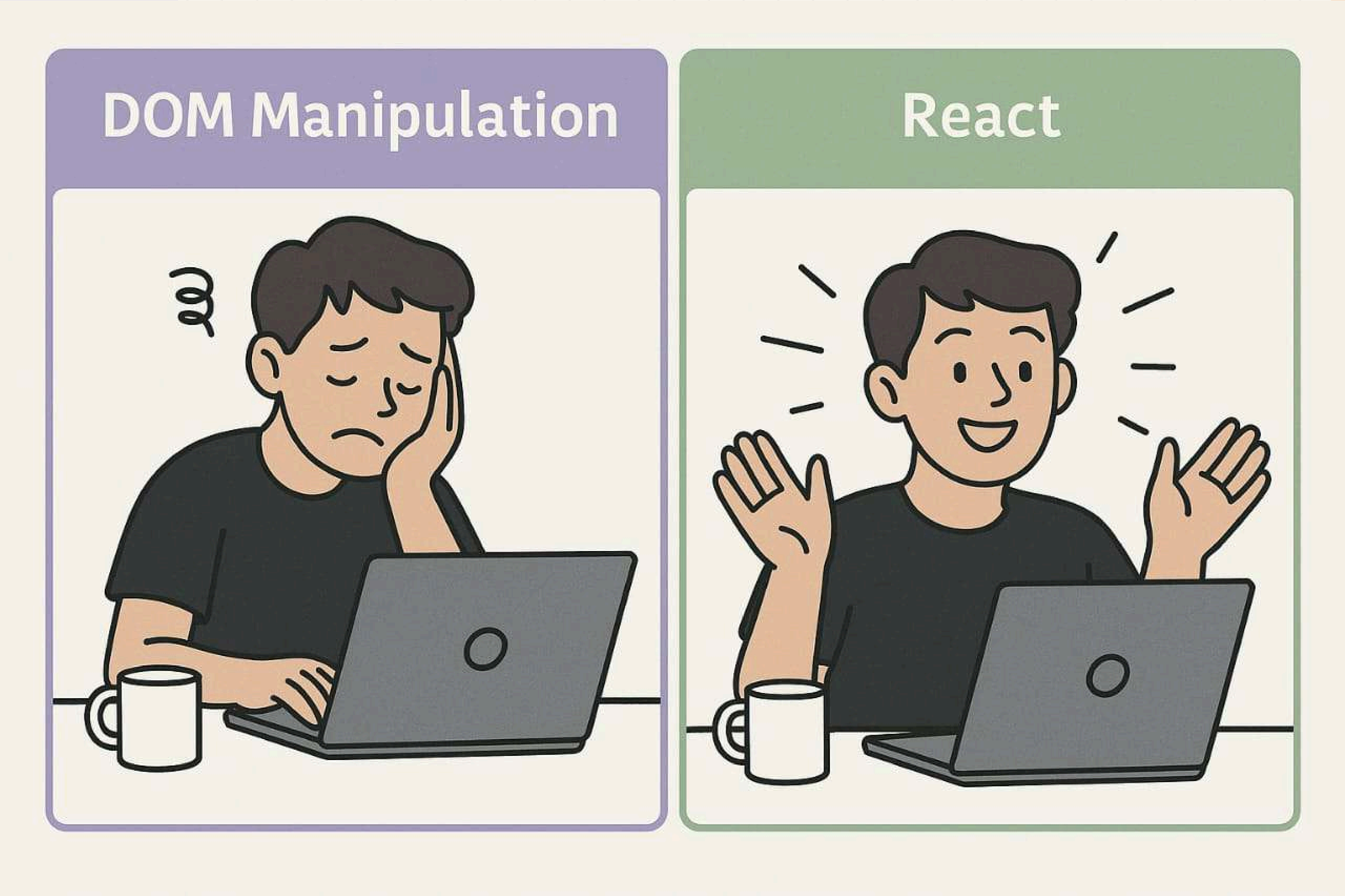
WHY REACT?

HTML CSS JS

STATIC DATA

REACT

DYNAMIC DATA


DOM Manipulation
React

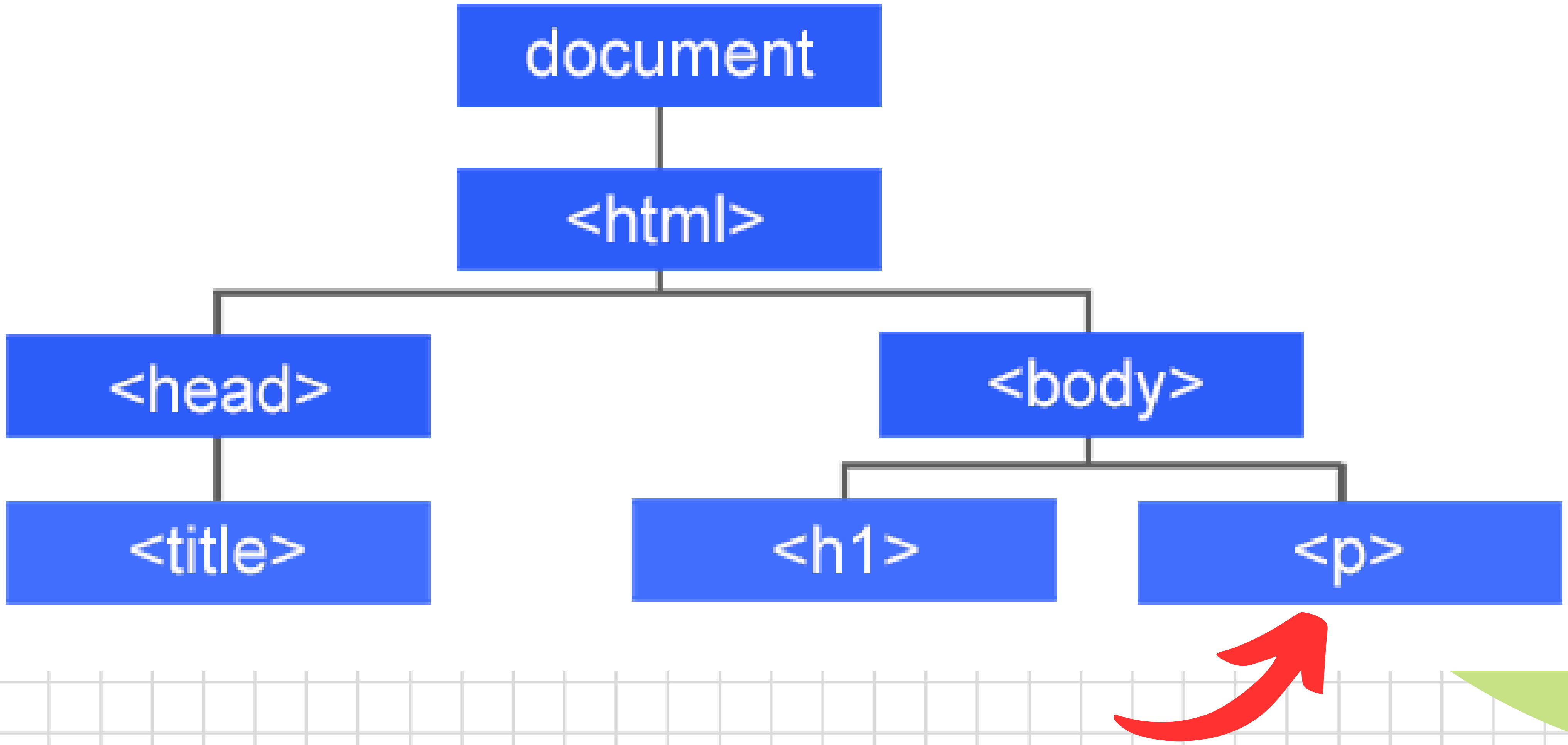
HOW REACT WORKS?

DOM VS VIRTUAL DOM

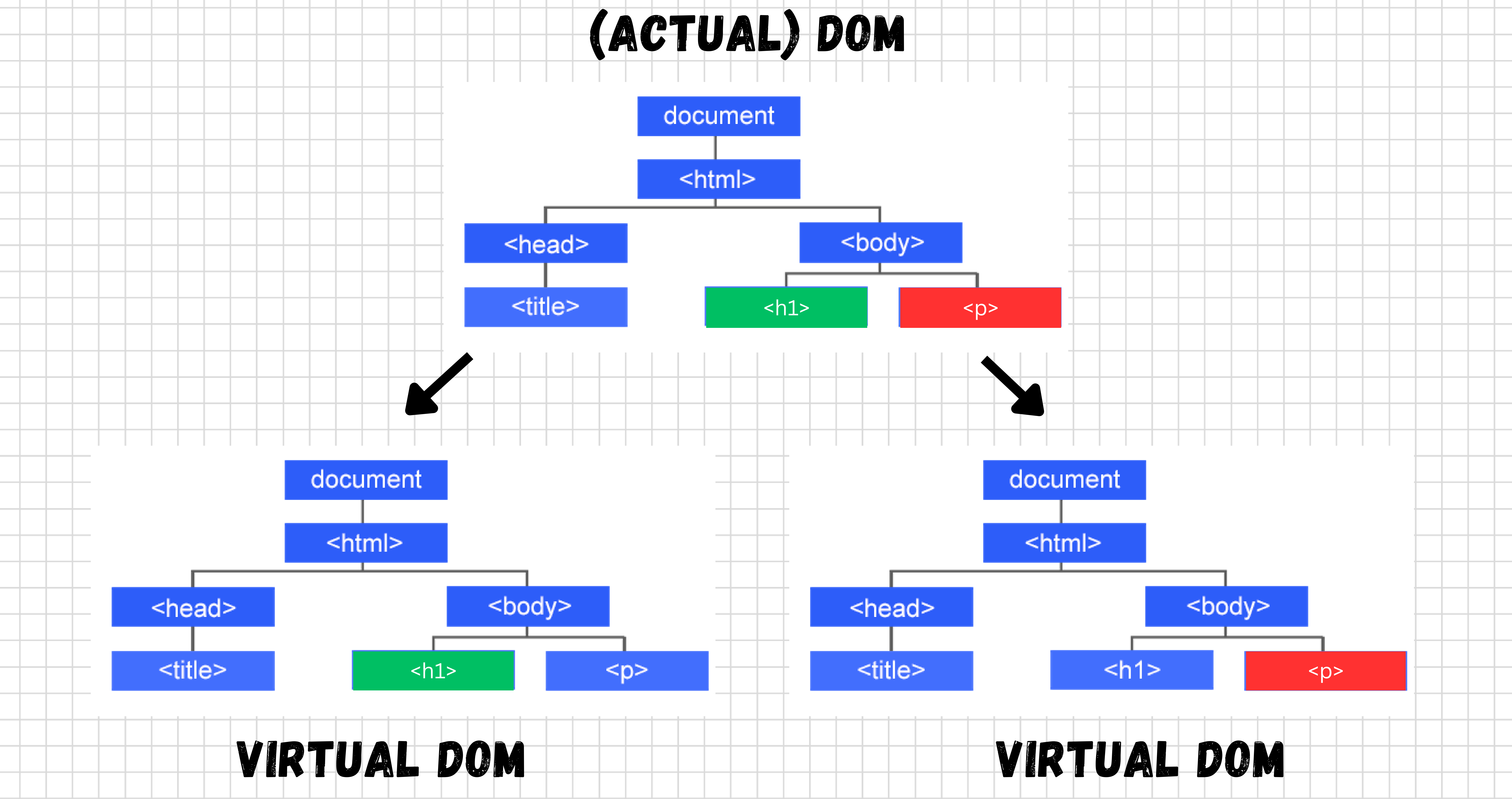
DOM - DOCUMENT OBJECT MODEL

```
<html>
    <head>
        <title>
    </head>
    <body>
        <h1></h1>
        <p></p>
    </body>
</html>
```

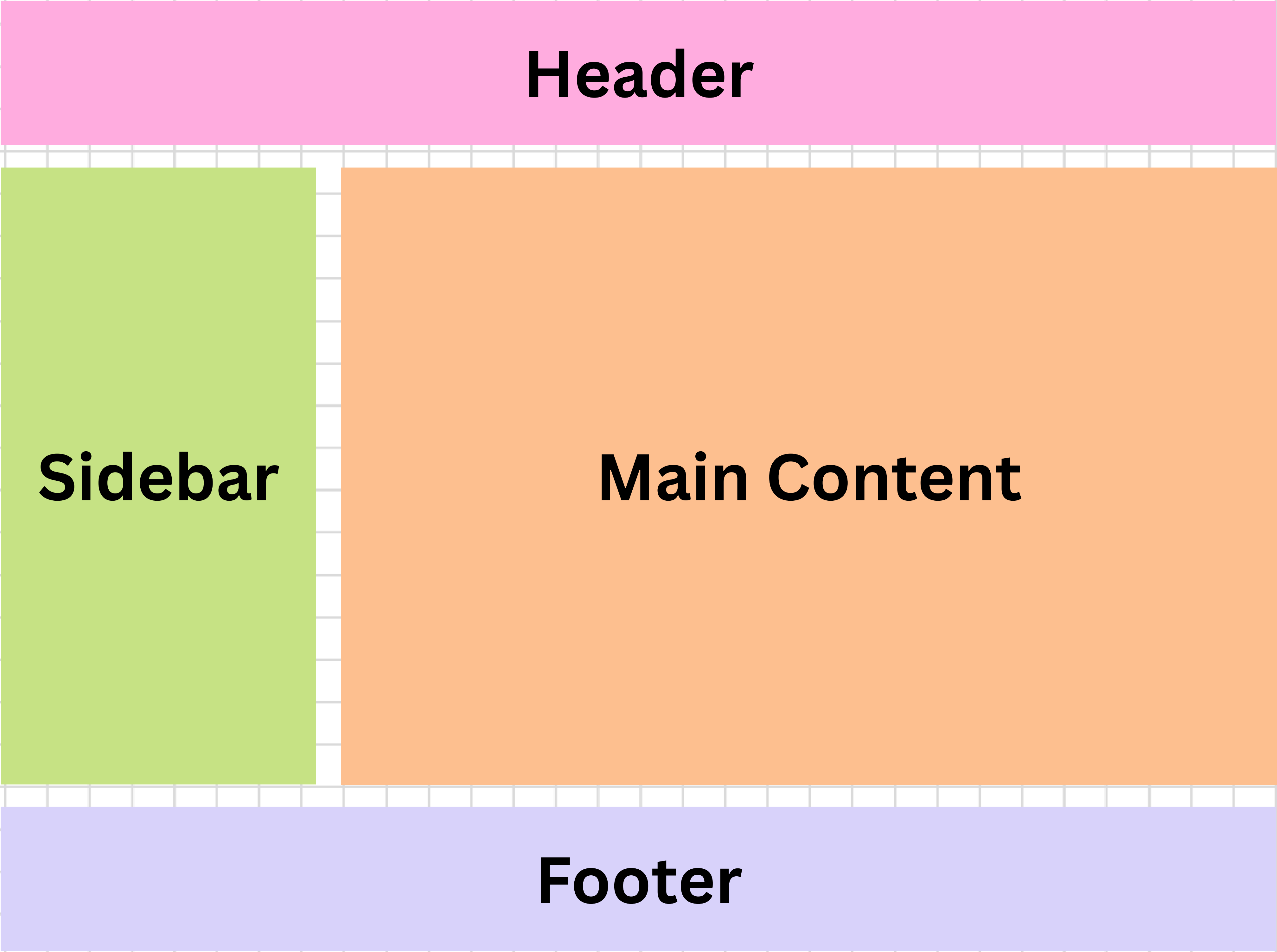
HTML

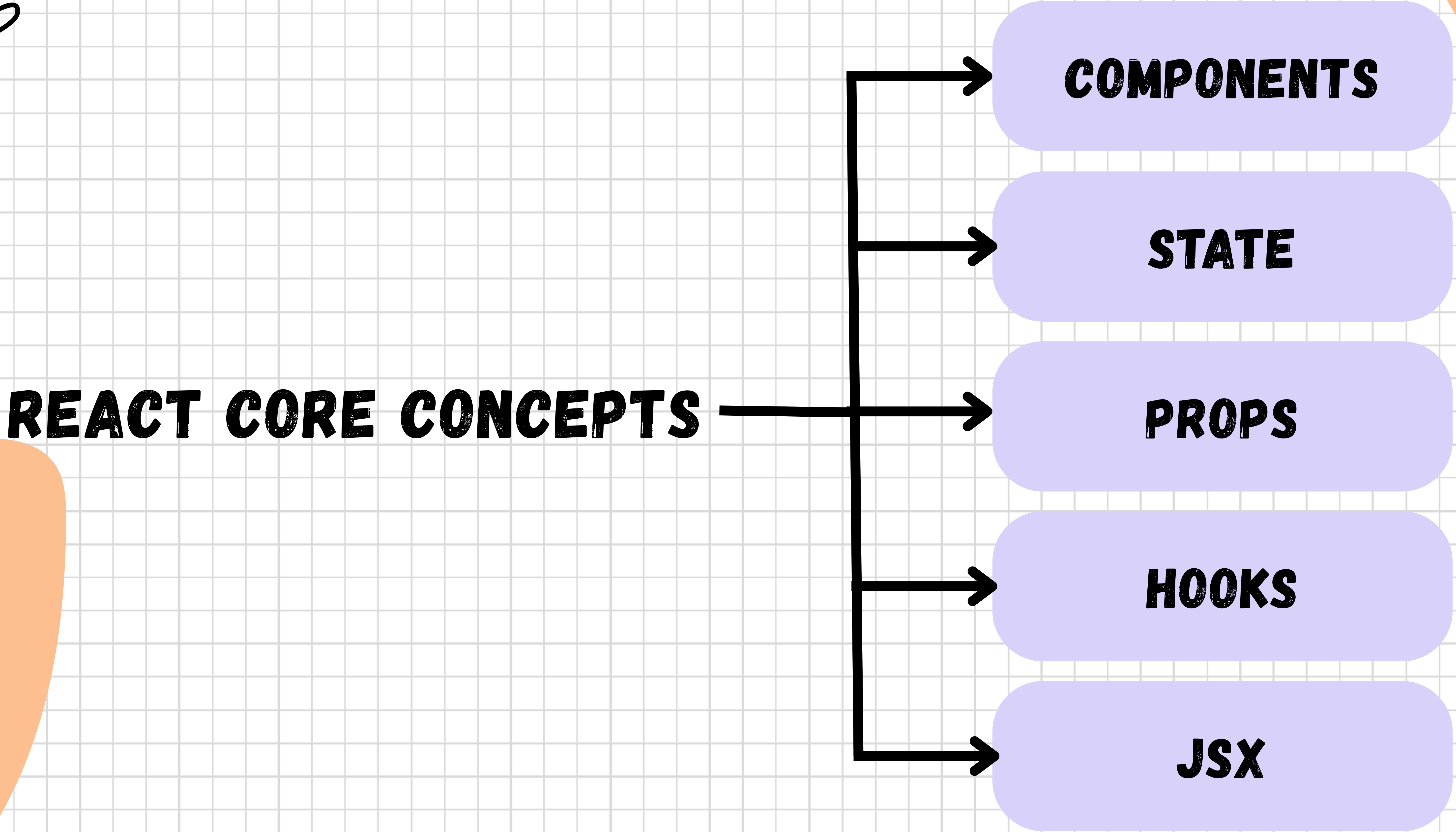


(ACTUAL) DOM


(ACTUAL) DOM
document
<html>
<head>
<body>
<title>
<h1>
<p>
document
<html>
<head>
<body>
<title>
<h1>
<p>
VIRTUAL DOM
document
<html>
<head>
<body>
<title>
<h1>
<p>
VIRTUAL DOM

COMPONENT BASED ARCHITECURE




REACT CORE CONCEPTS
COMPONENTS
STATE
PROPS
HOOKS
JSX

SETTING UP A REACT APP WITH VITE

JSX VS TSX

JSX

HTML-LIKE CODE INSIDE JAVASCRIPT

TSX

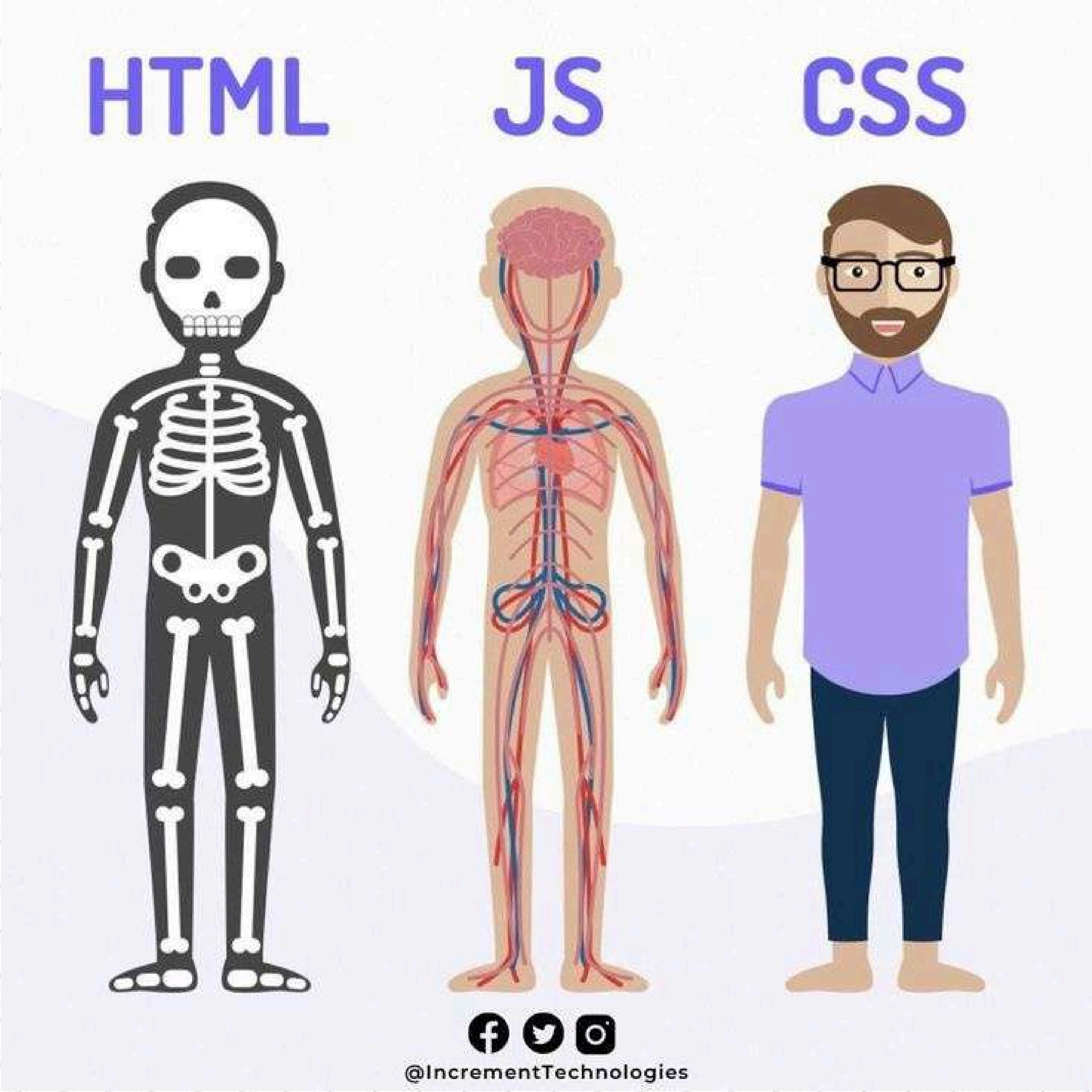
HTML-LIKE CODE INSIDE TYPESCRIPT

LETS BEGIN WITH JAVASCRIPT

Before we

Let’s Learn JS fundamentals...

What is js?



JavaScript Fundamentals

01

Variables Loops

Conditionals

02

Functions

Arrow func

Arrays

Objects

03

Callbacks

Promises

Async/await

04

try-catch

Import-

Export

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VARIABLES, LOOPS AND CONDÍTIONALS

JS

variables in js

var

variables can be re-declared and updated.A global scope varibale

let

variables cannot be re-declared but can be updated. A block scope varibale

const

variables cannot be re-declared or updated. A block scope variable

TYpes of Loops

For Loop

The for loop, for example, is handy for **iterating through arrays** or executing code a **specific number of times**. It has three parts: **initialization**, **condition**, and **iteration.**

While Loop

The while loop, is useful for executing code while a **specified condition** is true.

The while loop, runs while a **specified condition** is true.

Do while Loop

The do-while loop guarantees that the loop's code block **executes at least once** before the condition is checked.

FOr & While loop Example

For Loop

```
//for loop
for(let i=0;i<10;i++){
    console.log(i);
}
```

While

```
//while loop
let n = 12;
let i = 10;

while (i < n) {
    console.log(i);
    i++;
}
```

Condtionals

Conditional statements like if and else help us control the flow of our programs based on certain conditions. They're crucial for executing different blocks of code based on whether conditions are met."

Syntax:

```
if (condition) {
// Code to be executed if the condition is true
}
else {
// Code to be executed if the condition is false
}
```

FUNCTIONS

What and Why ?

Functions in **JavaScript** are self-contained **blocks of code** designed to **perform a specific task** or set of tasks

Functions allow you to **encapsulate** a piece of code, give it a name, and **reuse it throughout** your program.

```
function FunctionName(param1,param2){
    //function body
}
```

Arrow Functions

Concise Syntax: Arrow functions have a **shorter syntax** compared to regular functions, especially for single-expression functions.

Arrow (=>): Signifies the **creation** of an arrow function.

```
const functionName = (parameter1, parameter2) => {
  // Function body - code to be executed
  // It can use the parameters to perform operations

  return result; // returns a value
};
```

ARRAYS

WHAT IS AN

JavaScript arrays are versatile **data structures** with various functions and iteration methods that allow manipulation and traversal of array elements.

Arrays are **ordered collections** of values stored under a **single variable** name.

Arrays in JavaScript are created using square **brackets [ ] . Examples:**

```
const name = ["Gaurav","rohan","bhupesh","lokesh"]; //string
const number = [1,2,3,4,5]; //number
const mix = ["Akatsuki",1,true,null] //mixed
const empty = []; //null
```

Iteration

forEach():

```
//1) forEach
const anime=['Naruto','DemonSlayer','DeathNote'];
anime.forEach(animeS => { console.log(anime);
});
```

map():

```
//2) Map
const chars =anime.map(ch => ch + " anime");
console.log(chars);
```

filter():

```
const number = [1,2,3,4,5];
const even = number.filter((num)=> num%2 == 0);
console.log(even)
```

OBJECTS

Features

1. **Key-Value Pairs**

2. **Nested Objects**

3. **Dynamic Properties:** Properties can be added, modified, or deleted from objects dynamically.

4. **Accessing Properties:**
Dot notation: object.property
Bracket notation: object['property']

Syntax:

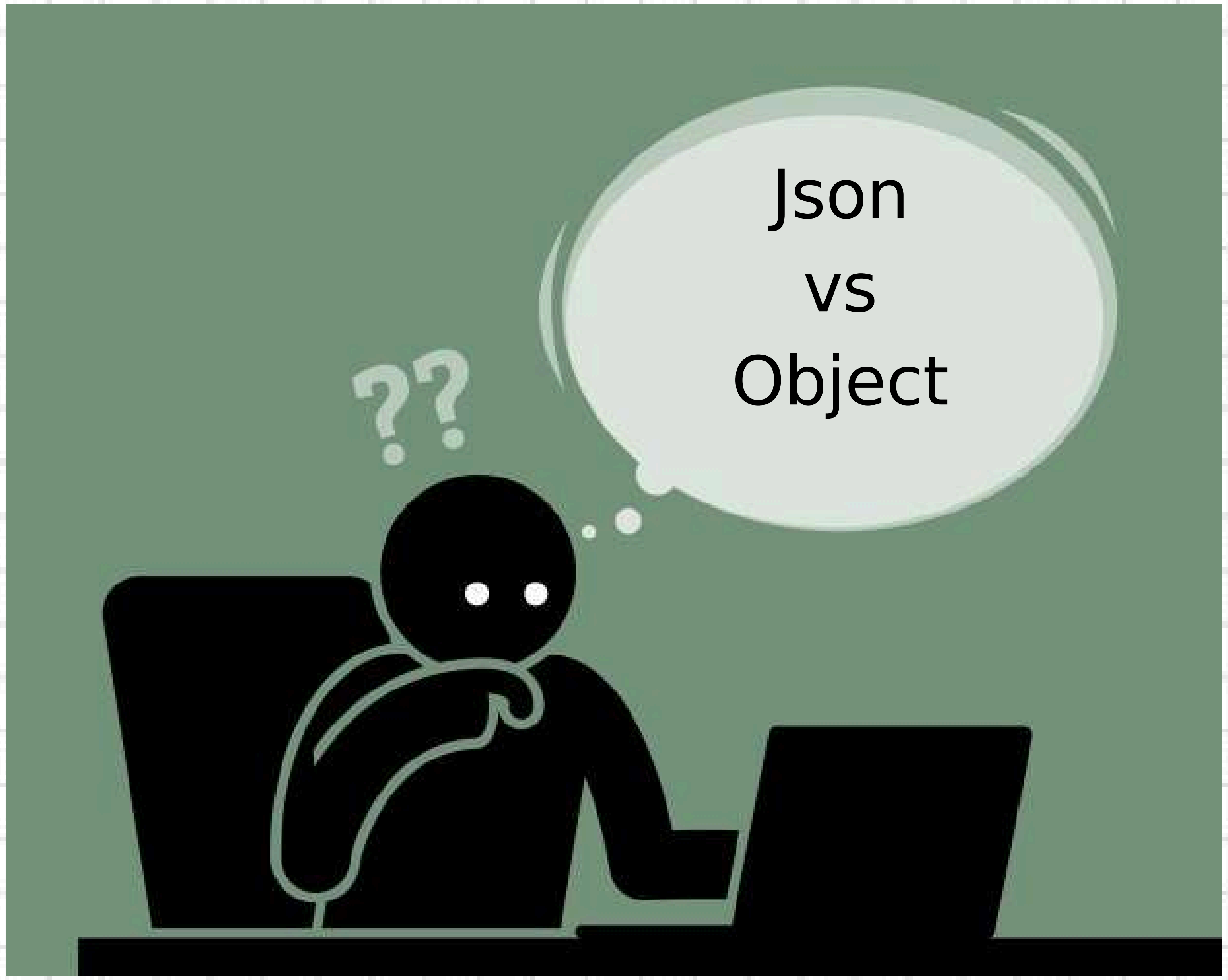
```
const object = {
   Key1 : Value_1,
   Key2 : Value_2


}


console.log(obj.a);
```

Object Methods:

```
const keys =Object.keys(obj); // returns the keys of the object
console.log(keys)
const values = Object.values(obj); // returns the values of the object
console.log(values)
const entries = Object.entries(obj); // returns the key-value pairs of the object
console.log(entries)
```

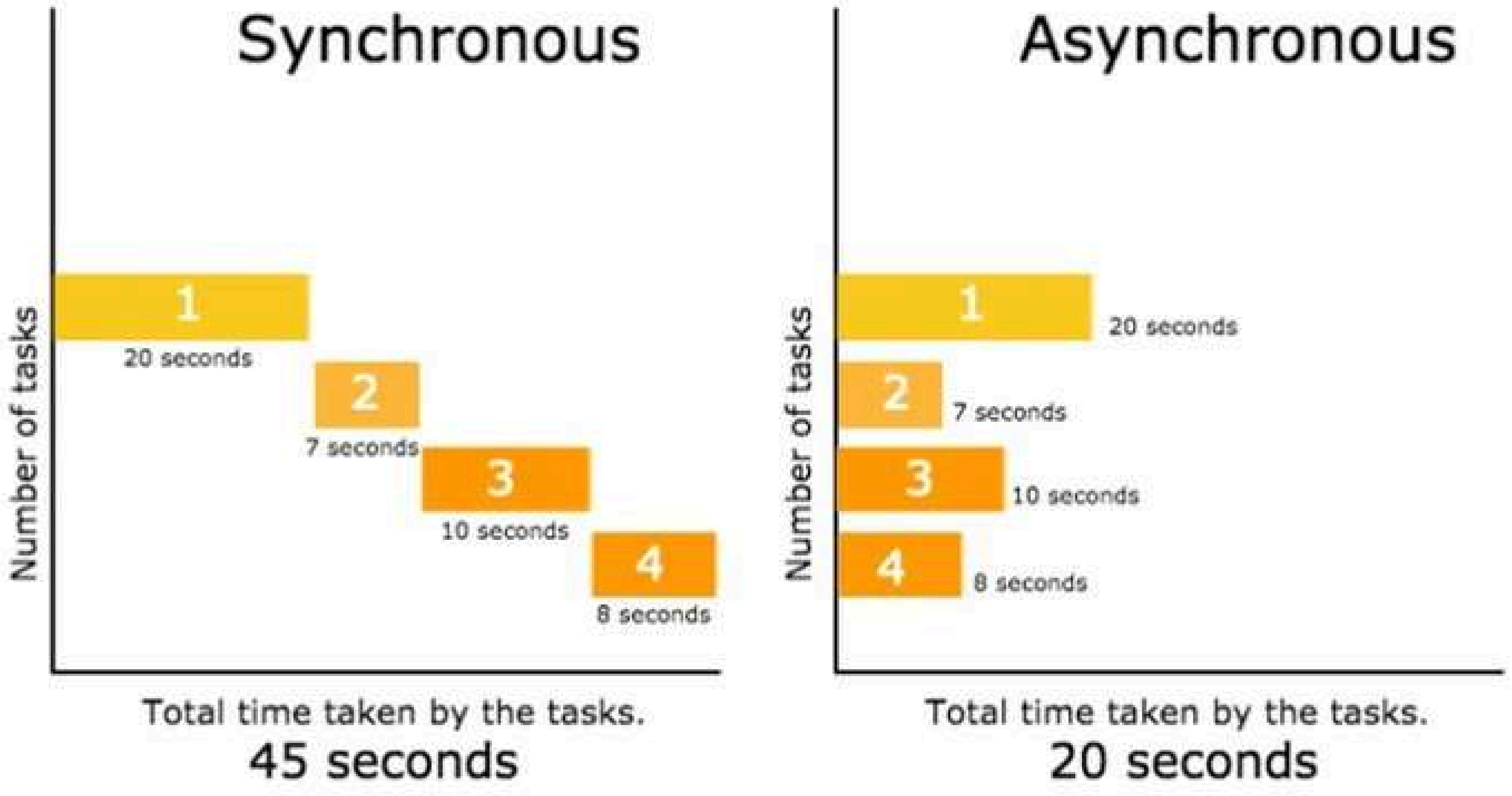
json vs object

Json :- JavaScript Object Notation

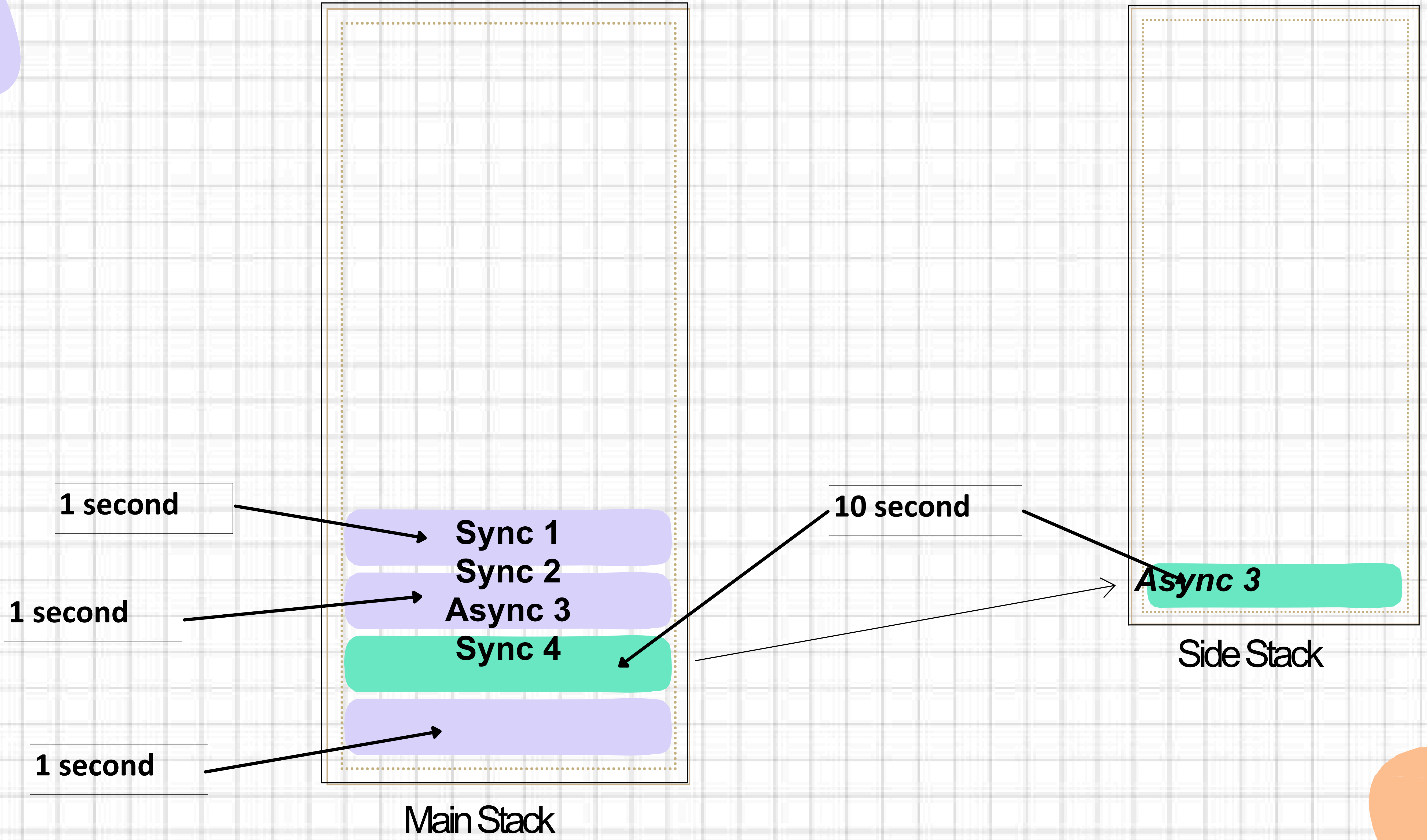


ASYNCHRONOUS JAVASCRIPT

synchronous vs asynchronous

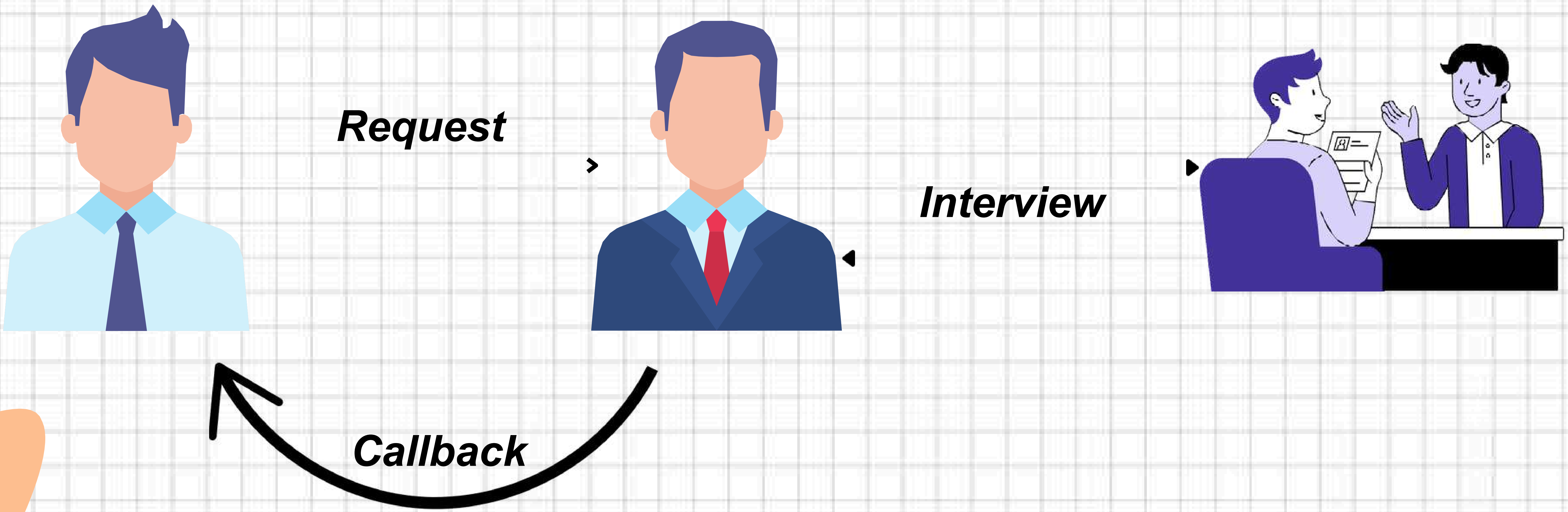


synchronous vs asynchronous



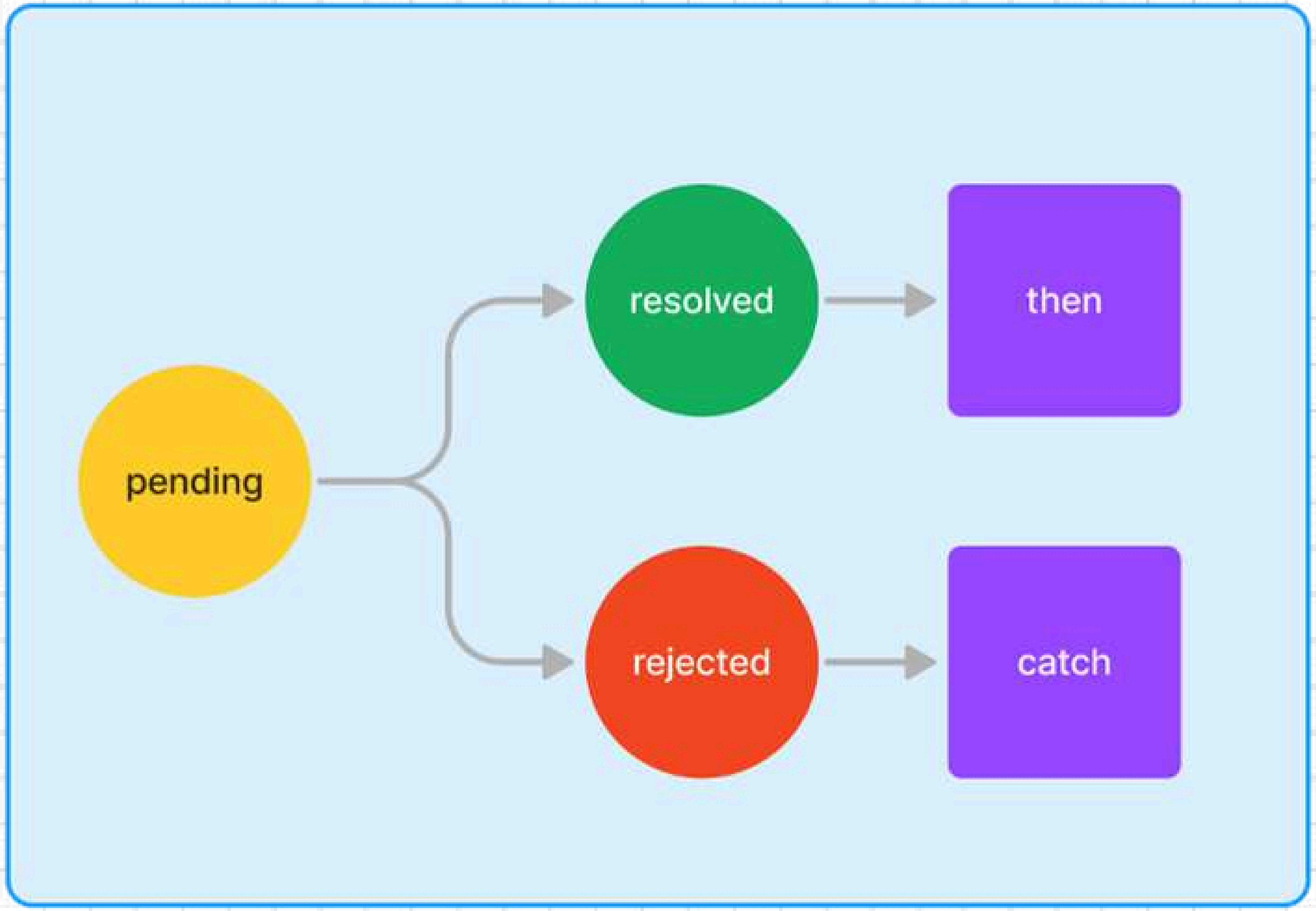
CALLBACKS

what is callback function ?



```
  function sayHello(name, callback) {
    console.log(`Hello, ${name}!`);
    callback();
}

function coding() {
    console.log("start coding!");
}
```

promises



```javascript
function makeCoffee(isBeansAvailable) {
  return new Promise((resolve, reject) => {
    setTimeout(() => {
      isBeansAvailable ? resolve("Coffee is ready! Enjoy! ") : reject(" No coffee today! Out of beans! ");
    }, 1500);
  });
}

makeCoffee(true) // Change to false for a sad day
  .then(console.log)
```

async/await

```
t messege=async()=>{
const data = await fetch("https
randomuser.me/api/");
console.log(data);

ege();
```

Easy to Use

Error Handling

Timeout

Fetch

```
ling
chData = ()=>{
https://randomuser.me/api/').then( (data)=
sole.log(data);
n data.json()
( (jsonData)=>{
le.log(jsonData);

 (error)=>{
le.log("error ala bho..");



()
```

Axios

```
fetching() {

ponce = await axios.get('https://rand
.log(responce.data);
ror) {
.log("Can not fetch data",error);
```

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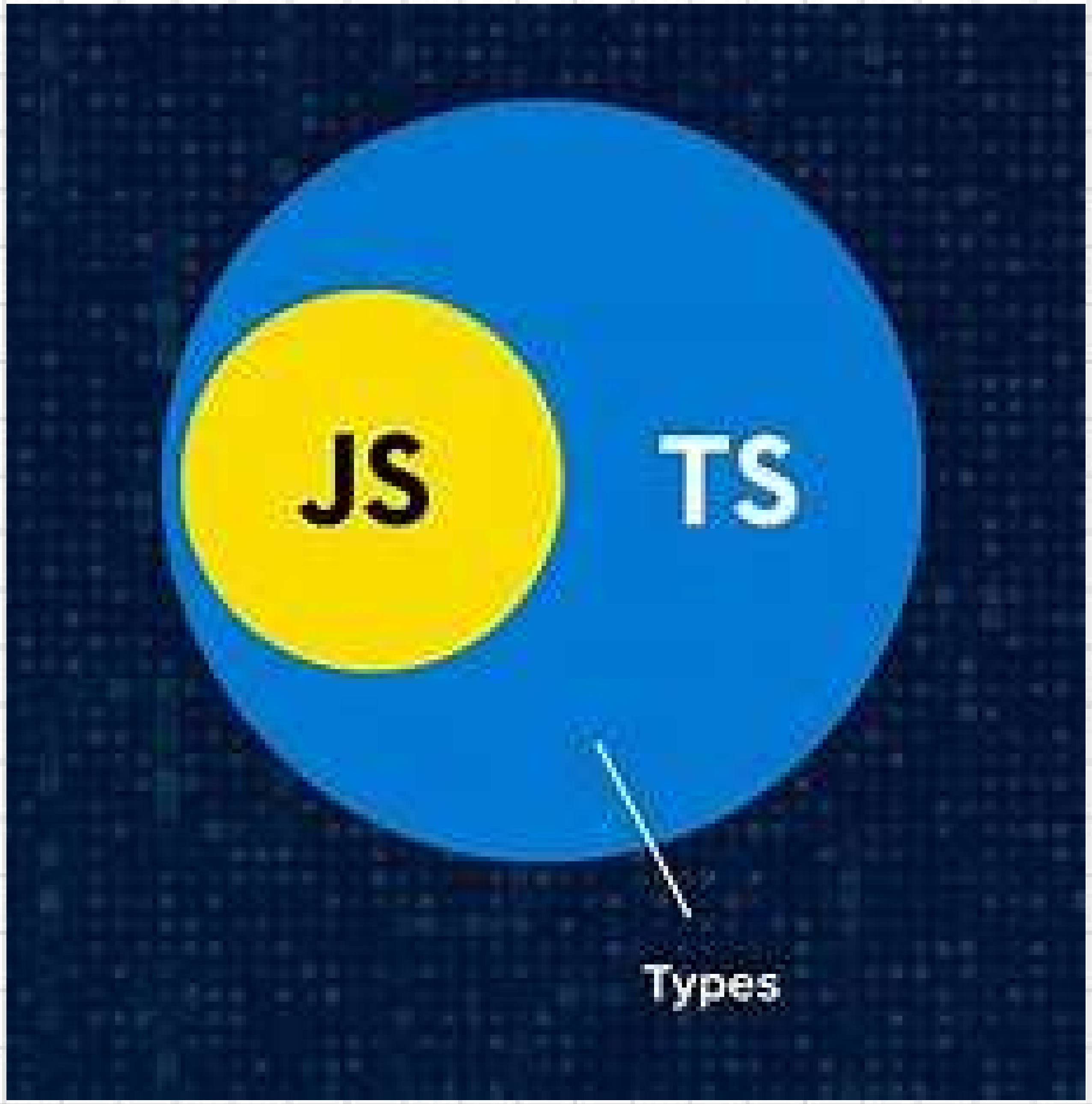
TS TypeScript

Choose TypeScript Over JavaScript?

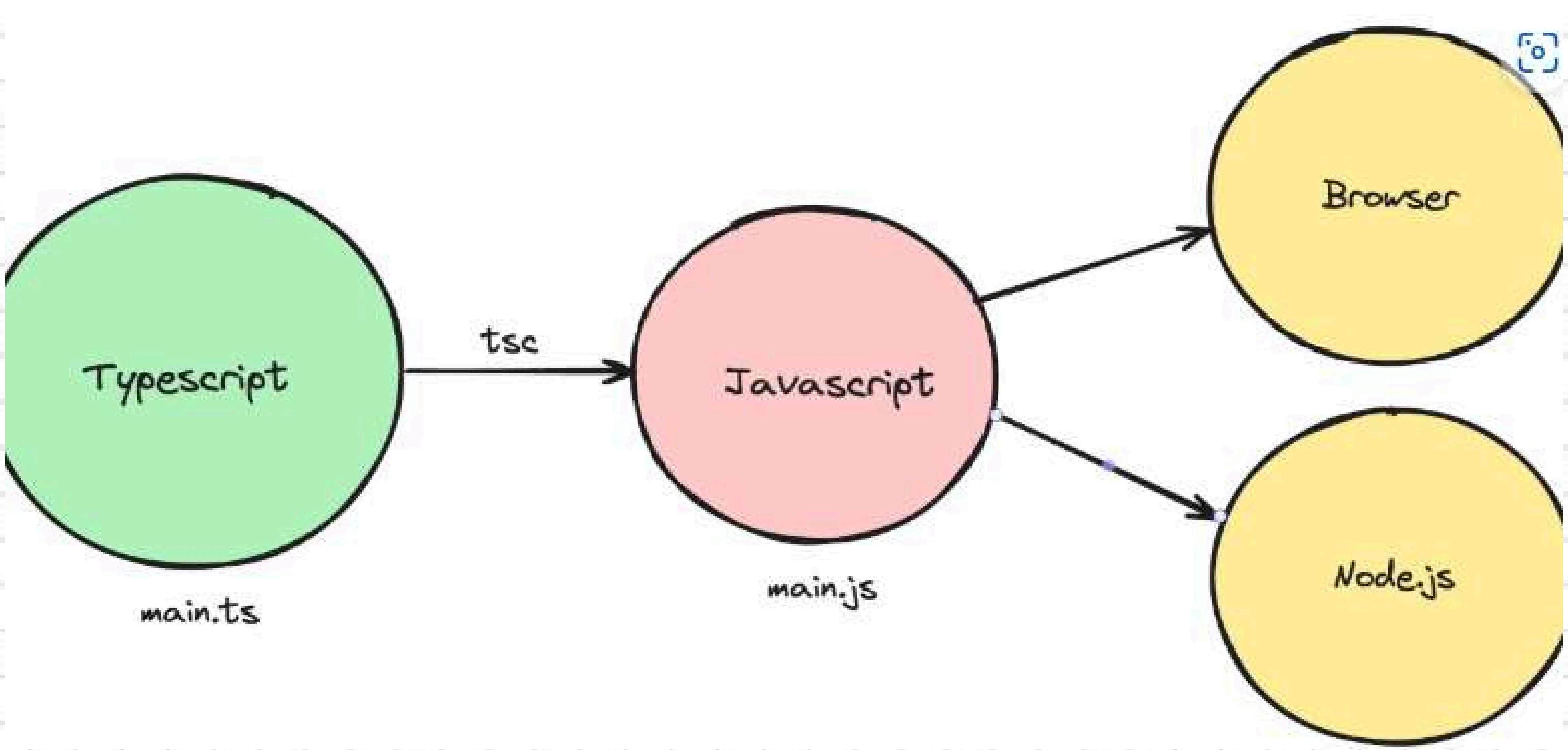
Static Typing

Compiled

Strict



How Typescript Run:



Installation:

1

```
-g typescript
```

2

```
-init
```

Let's Learn TS fundamentals...

01
Types

02
Array

03
Function

04
Interfaces

Number:

Represents both integers and floating-point numbers.

JS

```javascript
let age = 25;
console.log(age);

let price = 99.99;
console.log(price);
```

TS

```typescript
let age: number = 25;
console.log(age);

let price: number = 99.99;
console.log(price);
```

String:

Represents text values.

JS

```
let personname = "Akatsuki";
console.log(personname);
```

TS

```
let personname: string = "Akatsuki";
console.log(personname);
```

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Boolean:

Represents true or false.

JS

```javascript
let isActive = true;
console.log(isActive);
```

TS

```typescript
let isActive: boolean = true;
console.log(isActive);
```

Any:

Allows a variable to hold any type of value.

JS

```
let data = "Hello";
data = 42;
```

TS

```
let data: any = "Hello";
data = 42; //  Allowed
```

Null & Undefine:

Represents absence of value.

JS

```
let empty = null;
let notDefined = undefined;
console.log(empty);
console.log(notDefined);
```

TS

```
pty: null = null;
tDefined: undefined = undefine
e.log(empty);
e.log(notDefined);
```

Array:

Arrays store multiple values of the same type.

JS

```
2, 3, 4, 5]; // Array of numb
le", "Banana", "Mango"]; // A
s);
);
```

TS

```
[1, 2, 3, 4, 5];  // Arra
"Apple", "Banana", "Mango
```

Function:

Functions in TypeScript can have typed parameters and return types

JS

```
 add(a, b) {
urn a + b;

.log(add(5, 10)); // Output:
```

TS

```
 add(a: number, b: number): num
rn a + b;

log(add(5, 10));  // Output: 15
```

Interfaces:

Interfaces define the shape of an object, enforcing structure

JS

```javascript
let user = {
    name: "Jagruti",
    age: 20
};
console.log(user);
```

TS

```typescript
    name: string;
    age: number;
    isMarried?: boolean;  // Optional property
}

let user: Person = {
    name: "Jagruti",
    age: 20
};
console.log(user)
```

TAILWIND CSS

- UTILITY CLASSES
- PREDEFINE CILASS
- DESIGN EASILY & QUICKLY

VANILLA CSS VS TAILWIND CSS

BENEFITS OF TAILWIND

- FAST DEVELOPMENT
- CUSTOMIZABLE
- WORKS WITH ANY FRAMEWORK
- MOBILE-FIRST

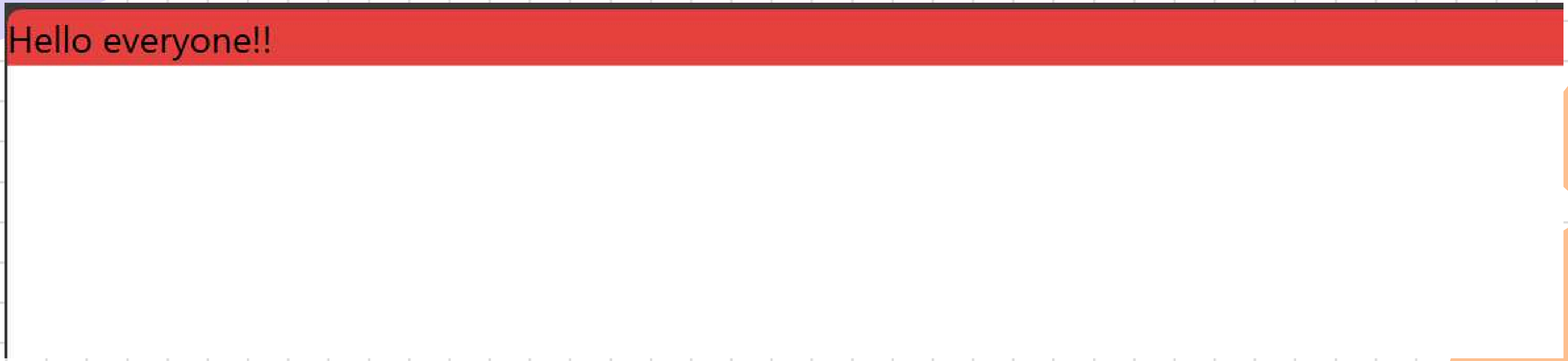
MOBILE FIRST

MEDIA QUERY COMMANDS IN TAILWIND

Breakpoint prefix	Minimum width	CSS
`sm`	40rem *(640px)*	`@media (width >= 40rem) { ... }`
`md`	48rem *(768px)*	`@media (width >= 48rem) { ... }`
`lg`	64rem *(1024px)*	`@media (width >= 64rem) { ... }`
`xl`	80rem *(1280px)*	`@media (width >= 80rem) { ... }`
`2xl`	96rem *(1536px)*	`@media (width >= 96rem) { ... }`

COLORS IN TAILWIND

```
return (
  <div className='bg-red-500'>
      Hello everyone!!
  </div>
)
```

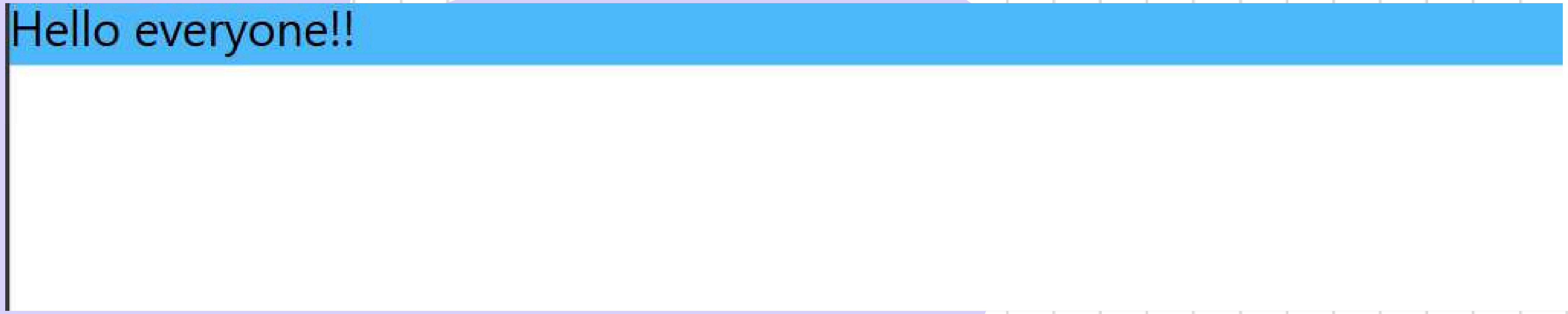


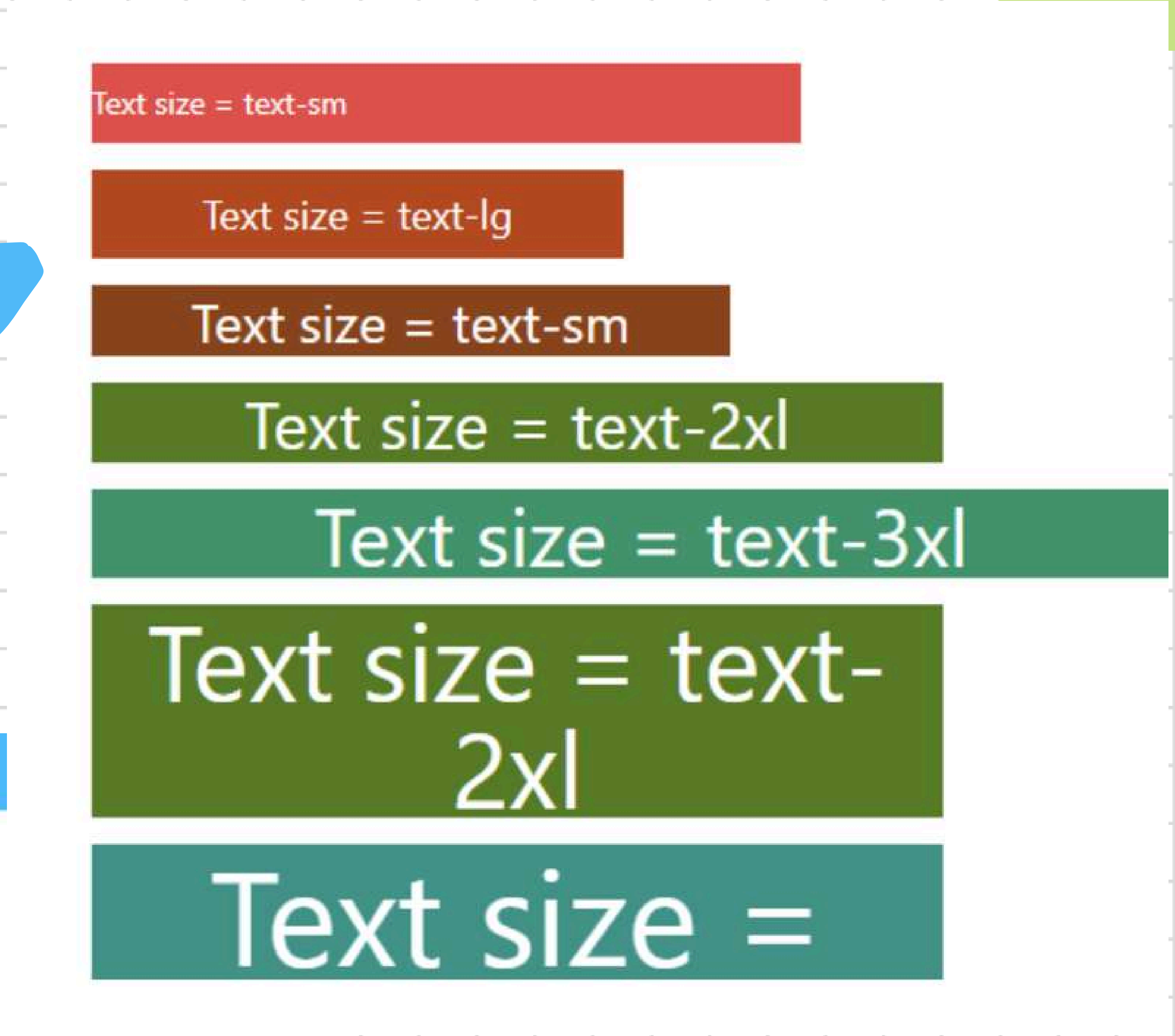
TEXT-COLOR IN TAILWIND

```
return (

    <div className='bg-black text-pink-500'>Welcome to Akatsuki</div>


)
```

Fonts in tailwind

```
  return (
    <div className='bg-sky-400 text-lg'>
        Hello everyone!!
    </div>
  )
}
```



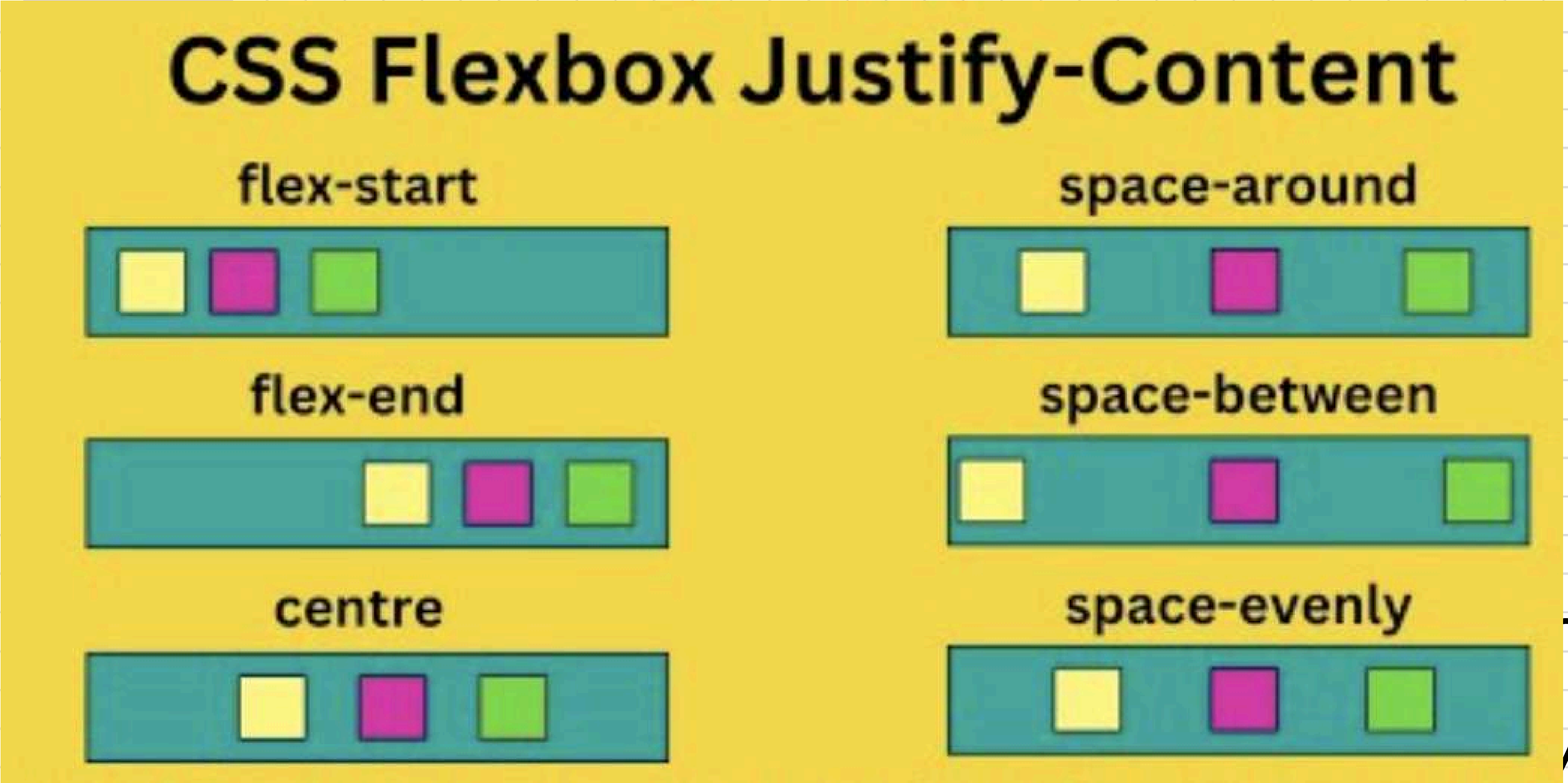


FLEX IN TAILWIND

```
return (
  <div className="bg-gray-100 p-4 flex flex-row items-center">
      <div className="bg-pink-500 p-4">Item 1</div>
    <div className="bg-green-500 p-4 ">Item 2</div>
    <div className="bg-yellow-500 p-4 ">Item 3</div>
    <div className="bg-red-500 p-4 ">Item 4</div>
    <div className="bg-blue-300 p-4 ">Item 5</div>
  </div>
)
```

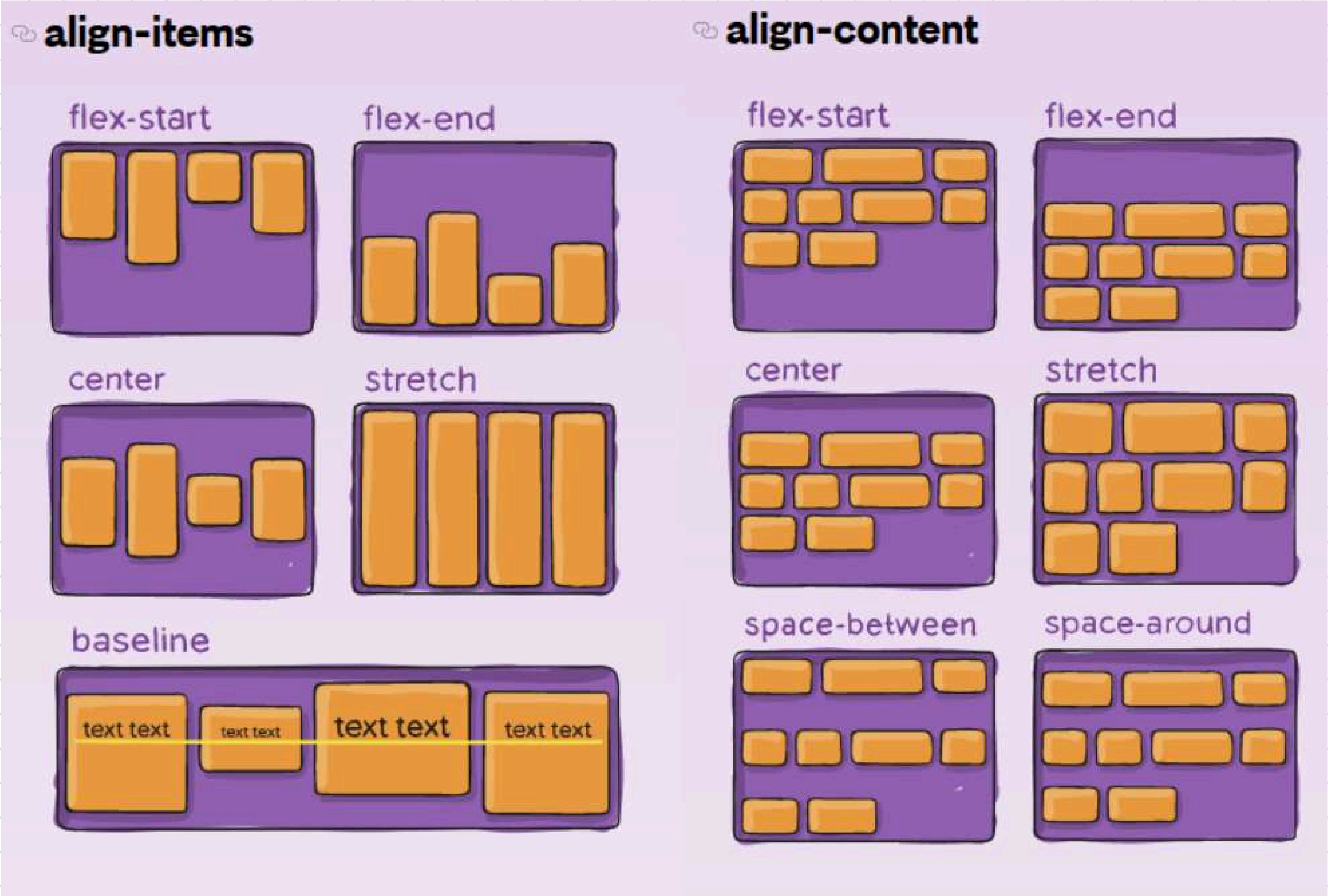
Important Flex Classes:

1. By default set to main axis



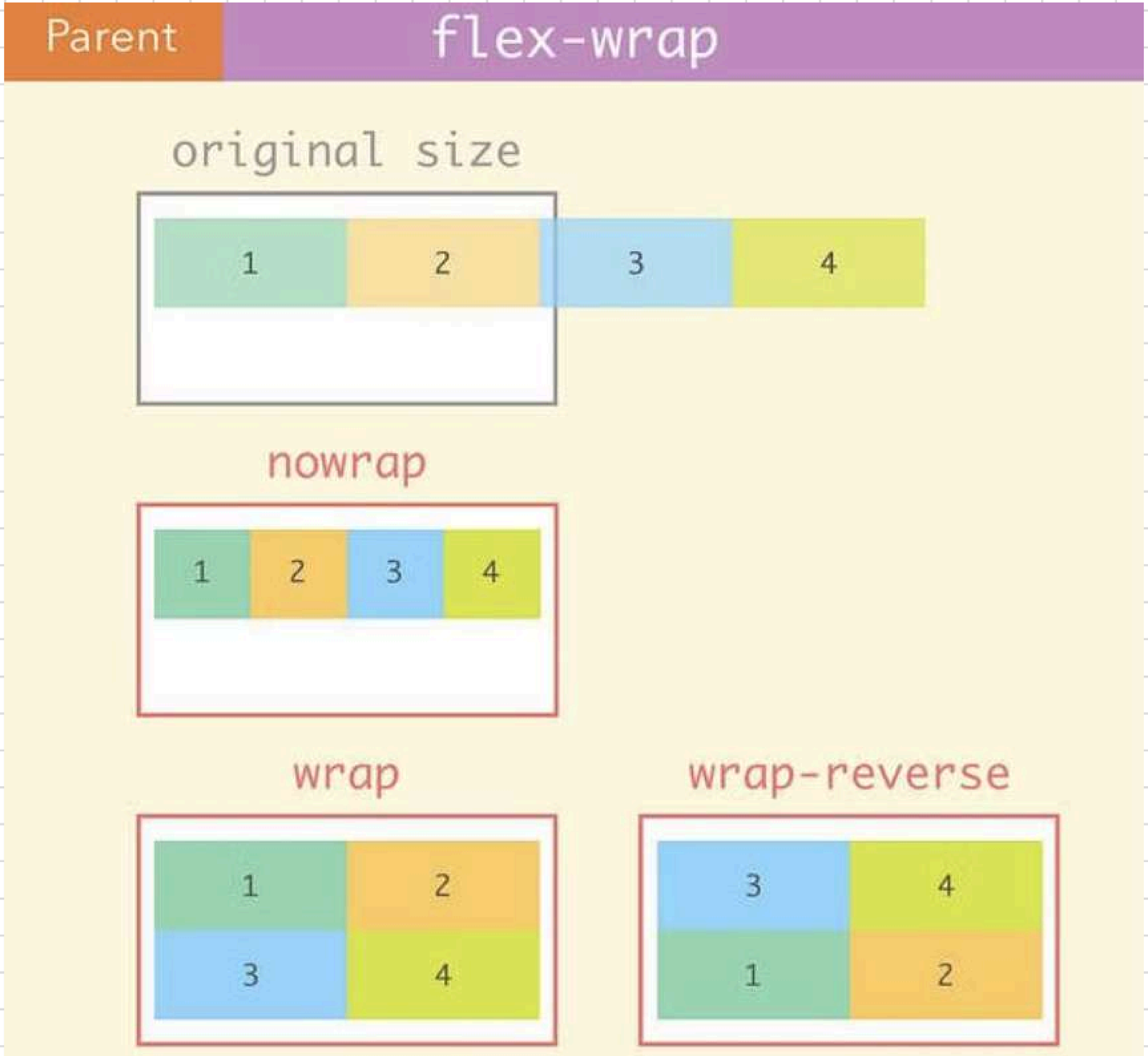
2.

By default set to cross axis



Only works when we set flex wrap

3.

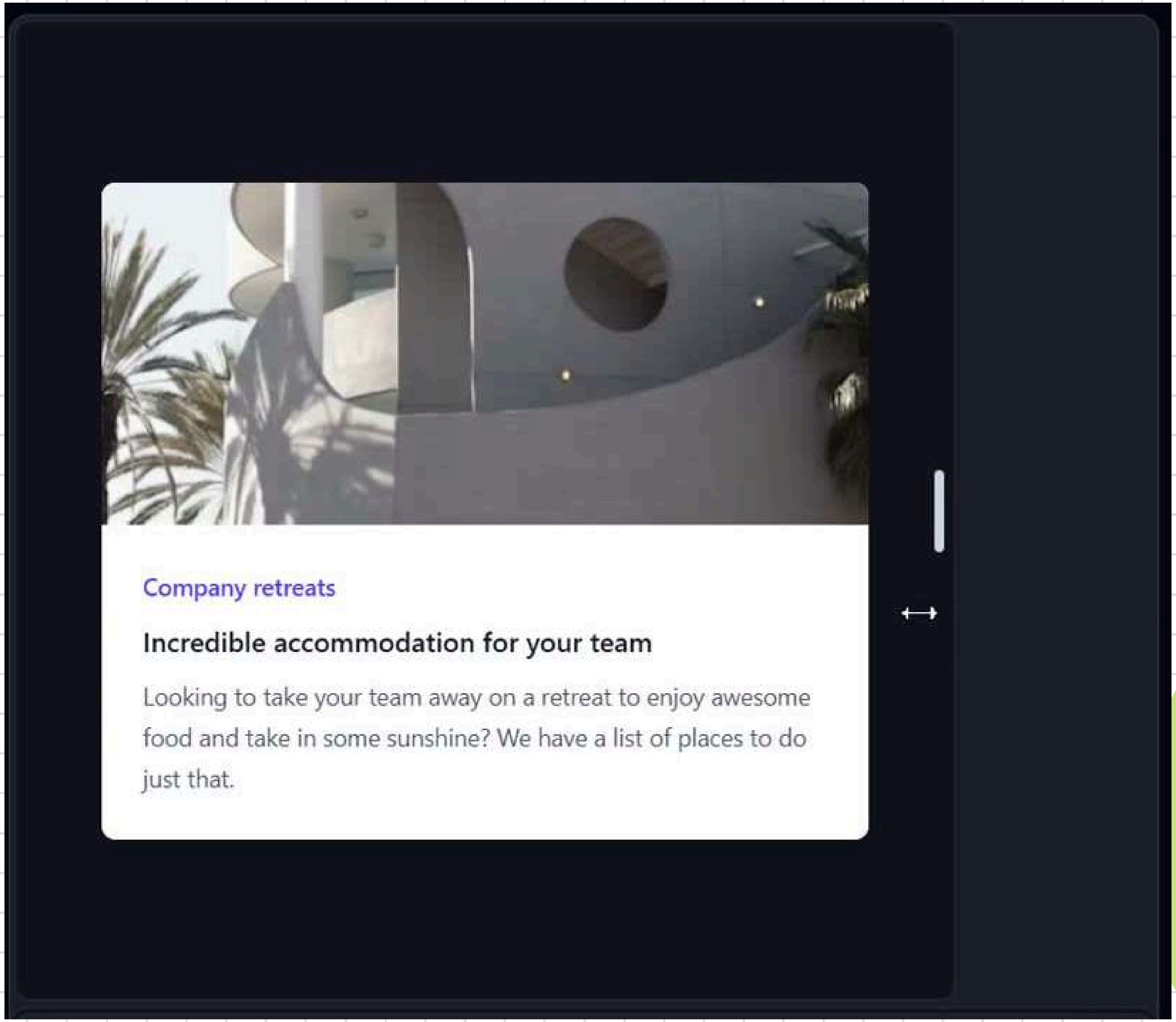

Parent
flex-wrap
original size
1
2
3
4
nowrap
1
2
3
4
wrap
1
2
3
4
wrap-reverse
3
4
1
2

GRID IN TAILWIND

```
return (
  <div className="grid grid-flow-row grid-cols-3 grid-col-3 gap-2 p-5 ">
  <div className="bg-red-500 text-white p-4 text-center rounded-lg">Grid Item 1</div>
  <div className="bg-green-500 text-white p-4 text-center rounded-lg">Grid Item 2</div>
  <div className="bg-yellow-500 text-white p-4 text-center rounded-lg">Grid Item 3</div>
  <div className="bg-blue-600 text-white p-4 text-center rounded-lg">Grid Item 4</div>
  <div className="bg-orange-300 text-white p-4 text-center rounded-lg">Grid Item 5</div>
  <div className="bg-violet-700 text-white p-4 text-center rounded-lg">Grid Item 6</div>
  <div className="bg-pink-500 text-white p-4 text-center rounded-lg">Grid Item 7</div>
  <div className="bg-lime-500 text-white p-4 text-center rounded-lg">Grid Item 8</div>
  <div className="bg-rose-500 text-white p-4 text-center rounded-lg">Grid Item 9</div>
</div>
)
```

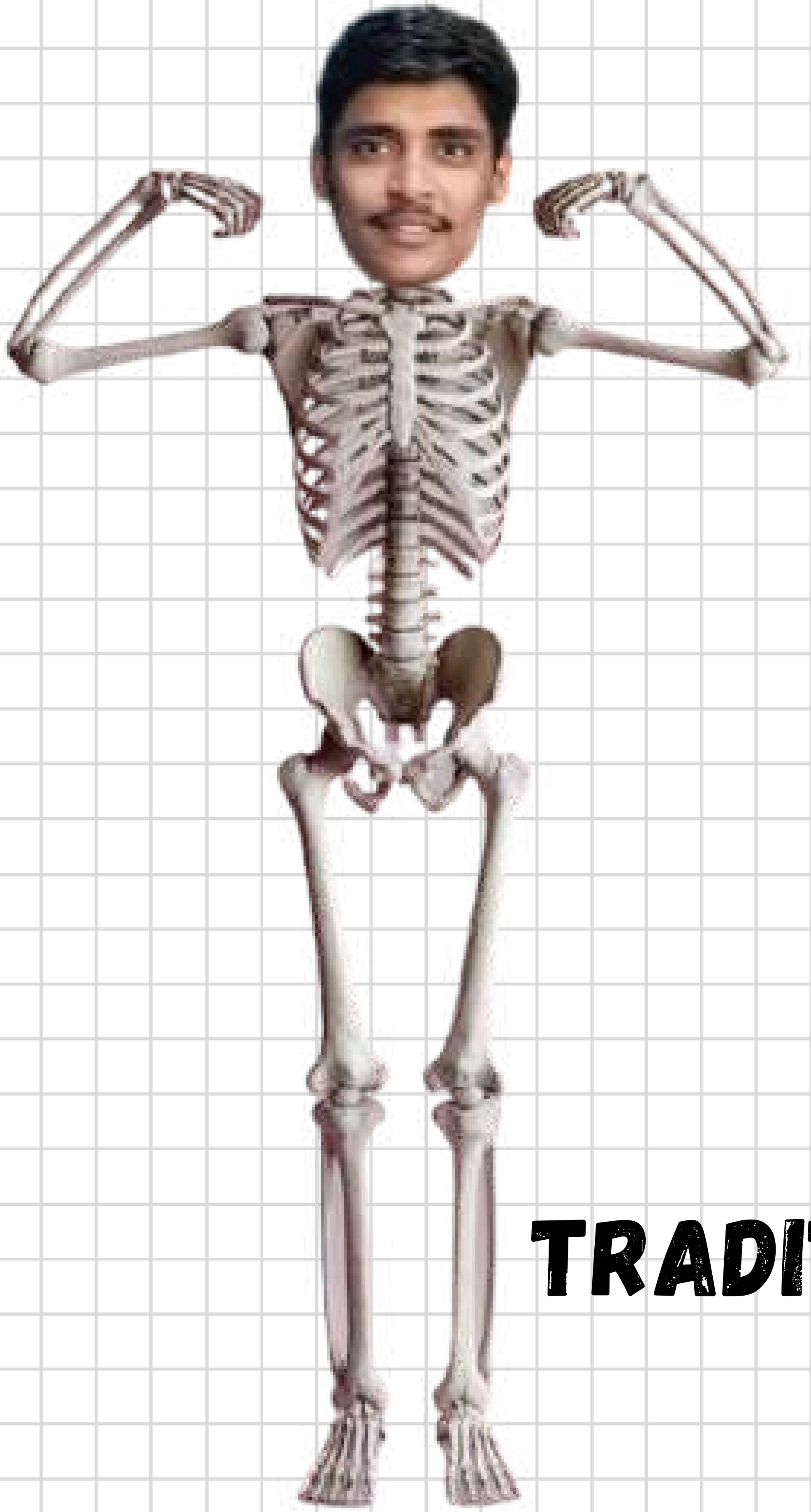
HOVER

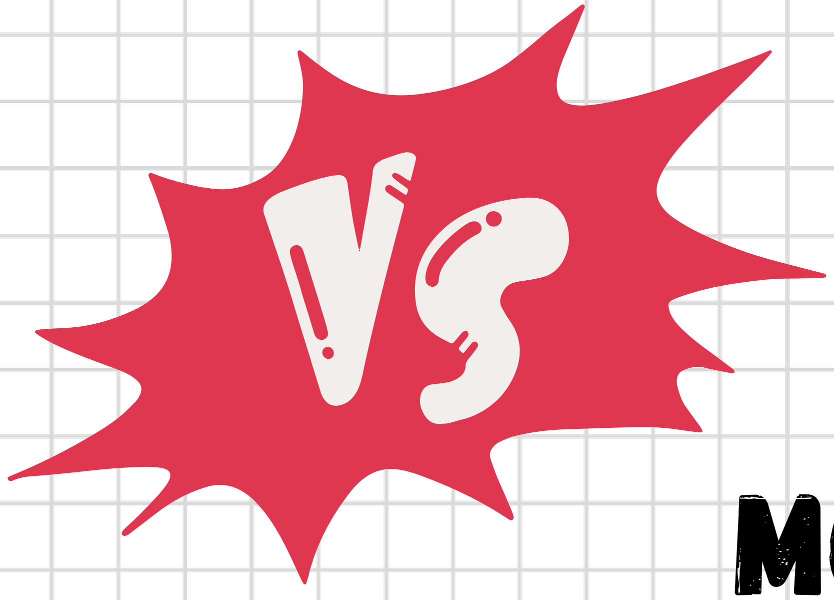
RESPONSIVENESS



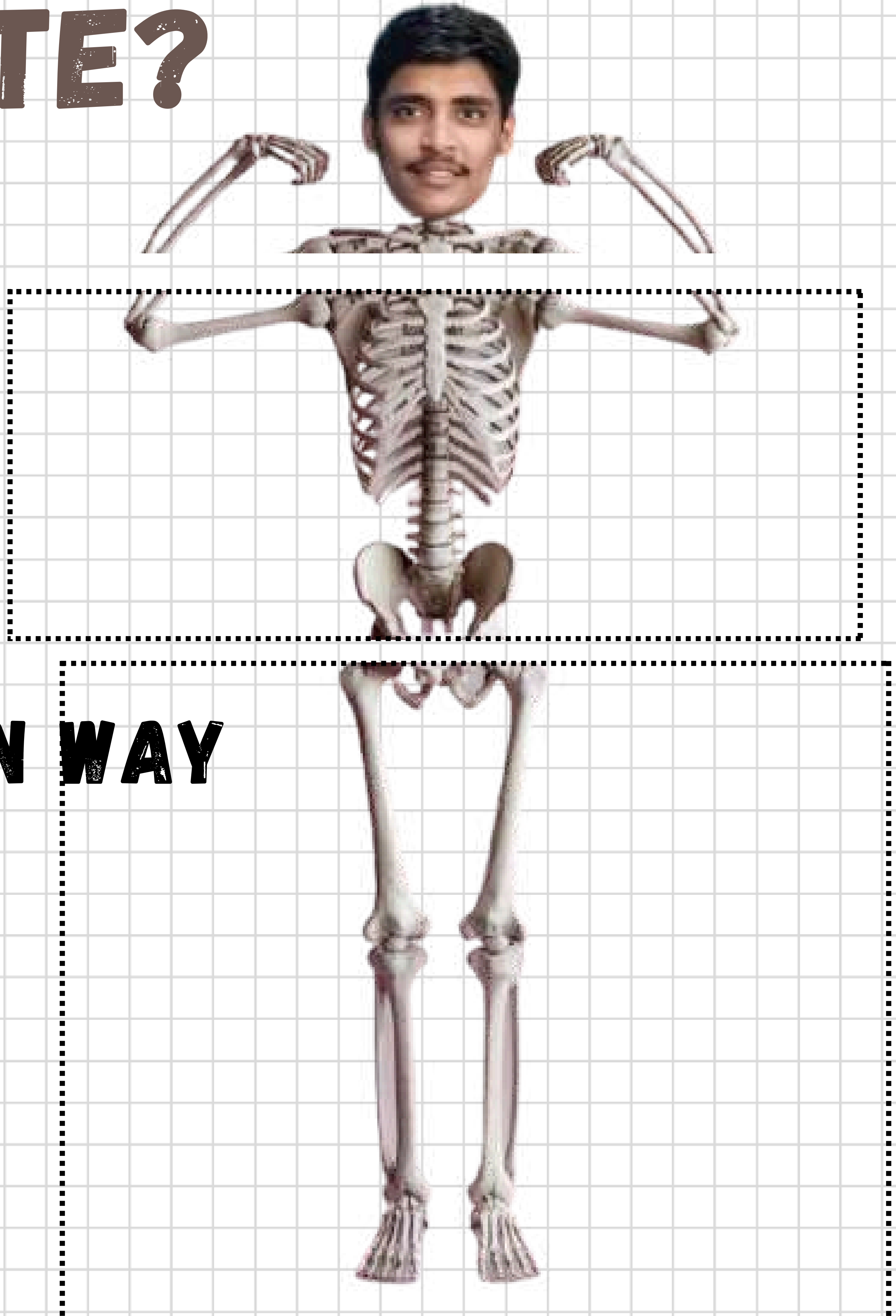
REACT PLATFORM

HOW WE CREATE WEBSITE?





TRADITIONAL WAY



MODERN WAY

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Create

9+

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Shorts

Subscriptions

You

All | Music | Dramedy | Podcasts | Thrillers | Ajay Devgn | Satire | Startup company | Mixes | Gaming | Electric vehicles | Superhero

Janavi Struggle For पुरण पोळी | Gudi Padwa Vlog 20

Dipraj Jadhav Edits

6.2K views • 2 days ago

Time Machine

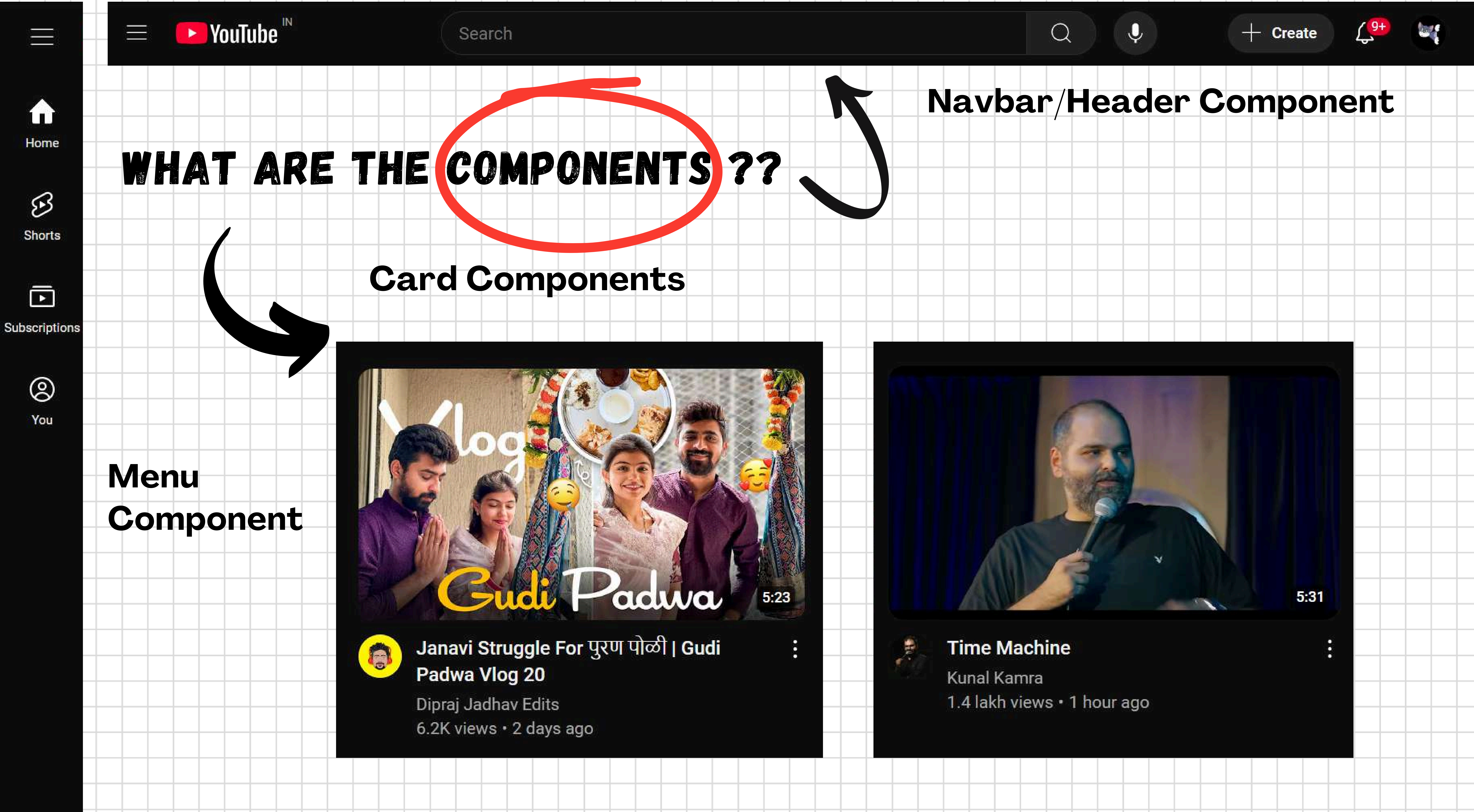
Kunal Kamra

1.4 lakh views • 1 hour ago

Natarang Ubha | Ajay-Atul Live Performance | MahaShivRatri 2025

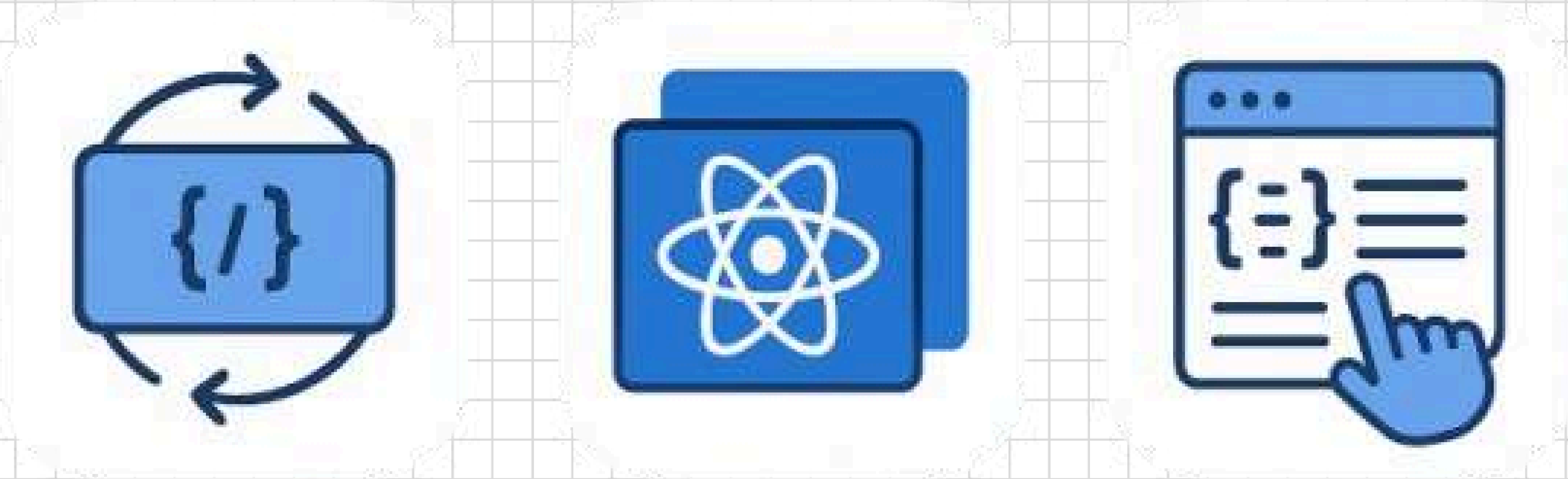
Zee Music Marathi

2.6 lakh views • 4 weeks ago



WHAT ARE THE COMPONENTS ??
Navbar/Header Component
Card Components
Menu Component
YouTube IN
Search
Create
9+
Home
Shorts
Subscriptions
You
Gudi Padwa
Vlog
5:23
Janavi Struggle For पुरण पोळी | Gudi Padwa Vlog 20
Dipraj Jadhav Edits
6.2K views • 2 days ago
5:31
Time Machine
Kunal Kamra
1.4 lakh views • 1 hour ago

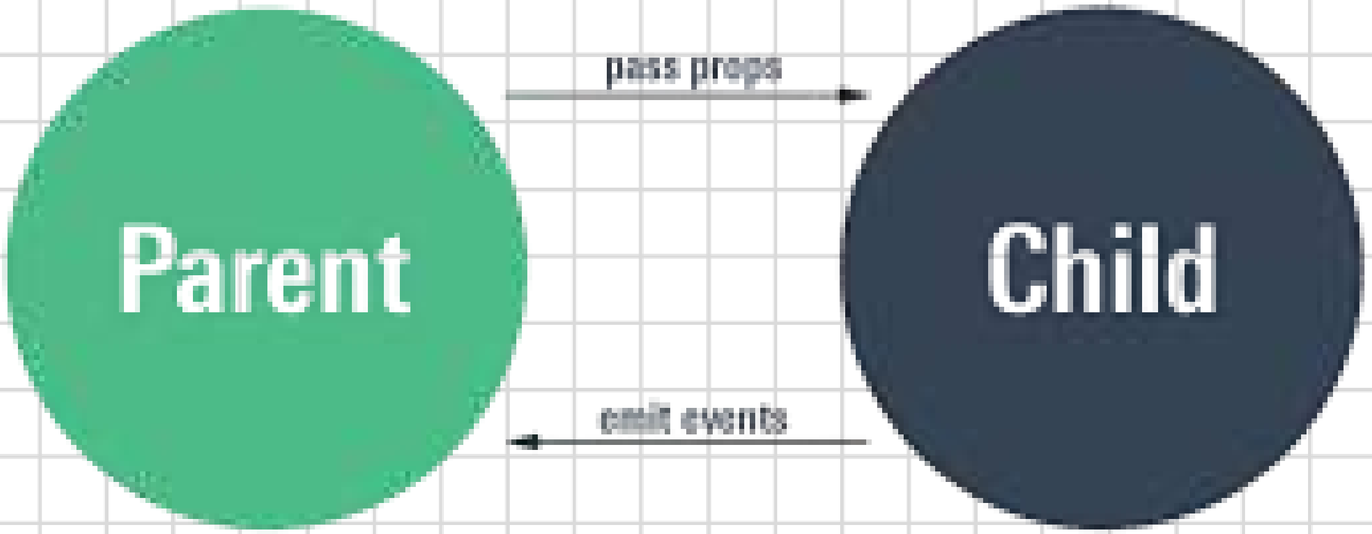
React Component Benefits



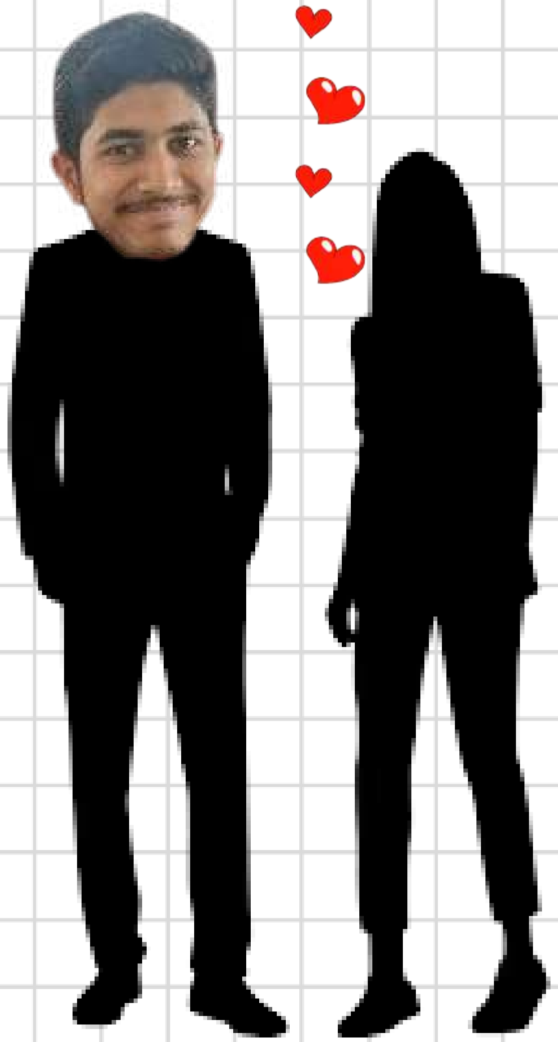
Reusability   Component-Based   Declarative

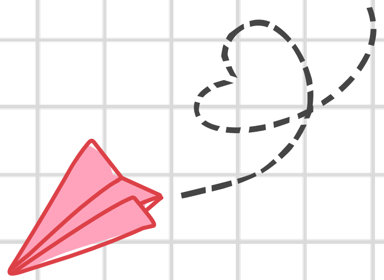
PROPS

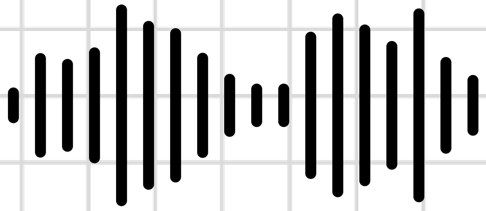
Props is a special keyword in React that stands for properties and is used for passing data from one component to another



PROPS



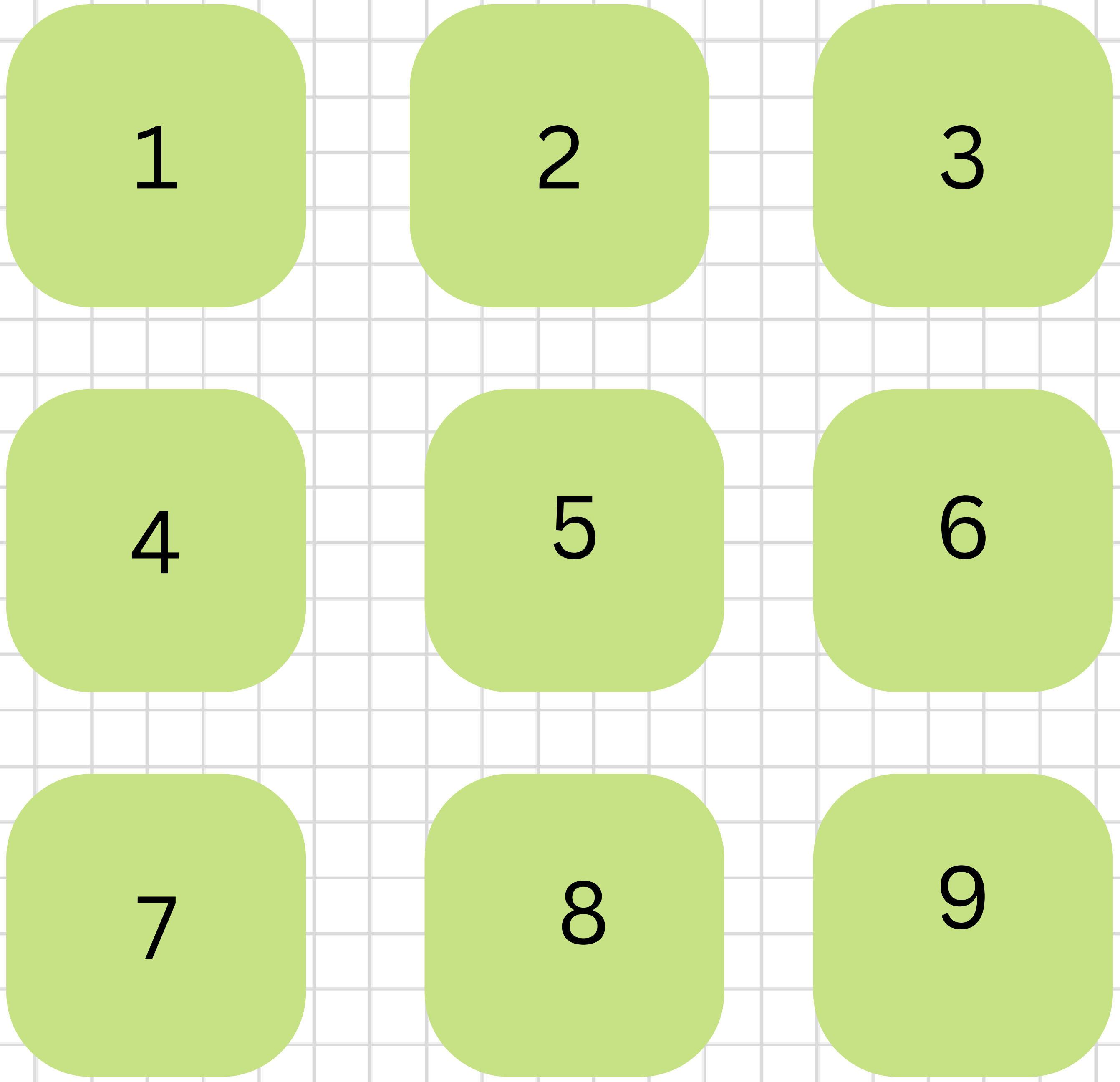




MAPPING

The maps are used for traversing or displaying the list of similar objects of a component

Without Mapping



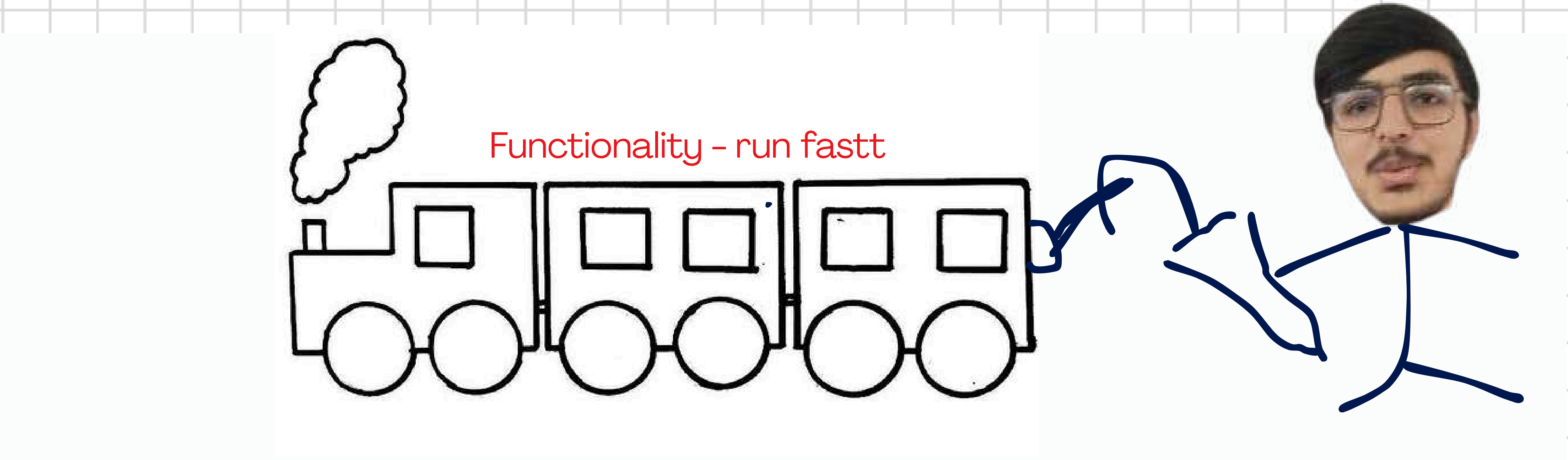
With Mapping

[1,2,3,4,5,6,7,8,9].map(()=>{

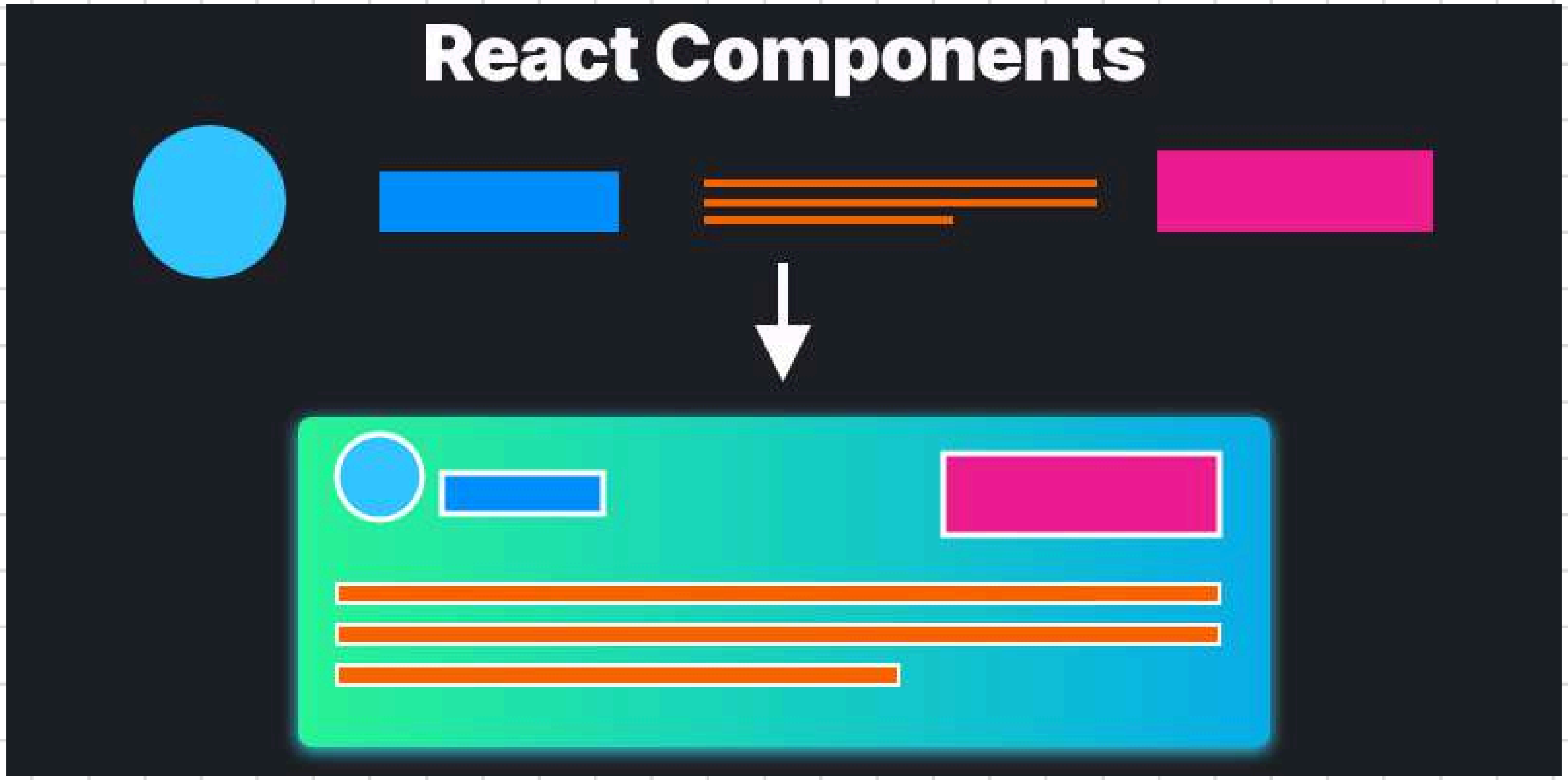
}

WHAT IS HOOKS ?


React
HOOK
Functionality


Functionality - run fastt

useState Hook



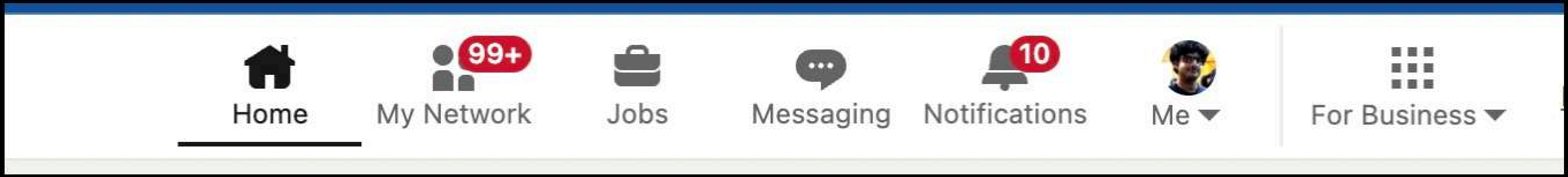
State

WHAT IS STATE ?

COMPONENT

```
Untitled-1

{
  topbar: {
    home: 0,
    myNetwork: "99+",
    jobs: 0,
    messaging: 0,
    notificaitons: 10
  }
}
```



Initial state

```
const [state, setState] = useState(0)
```

State variable

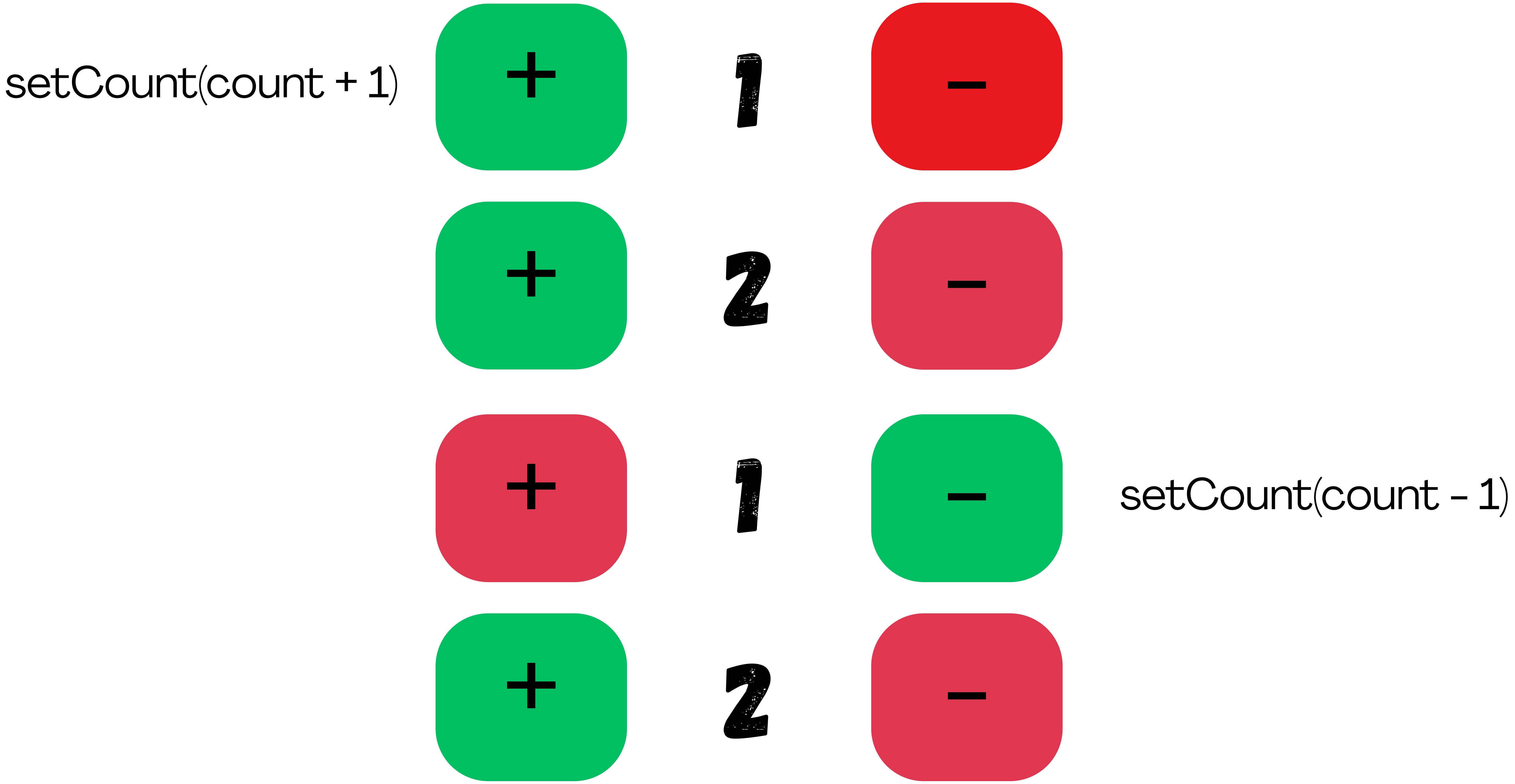
State function

State = 0

SetState(15)

State = ~~0~~ 15

const [count, setCount] = useState(0)



IPL · Sun, 23 Mar

MI	0/0 (20)	0/0 (20)	CSK

IPL · Sun, 23 Mar

MI	155/9 (20)	158/6 (19.1)	CSK

Routing?

ROUTES ?

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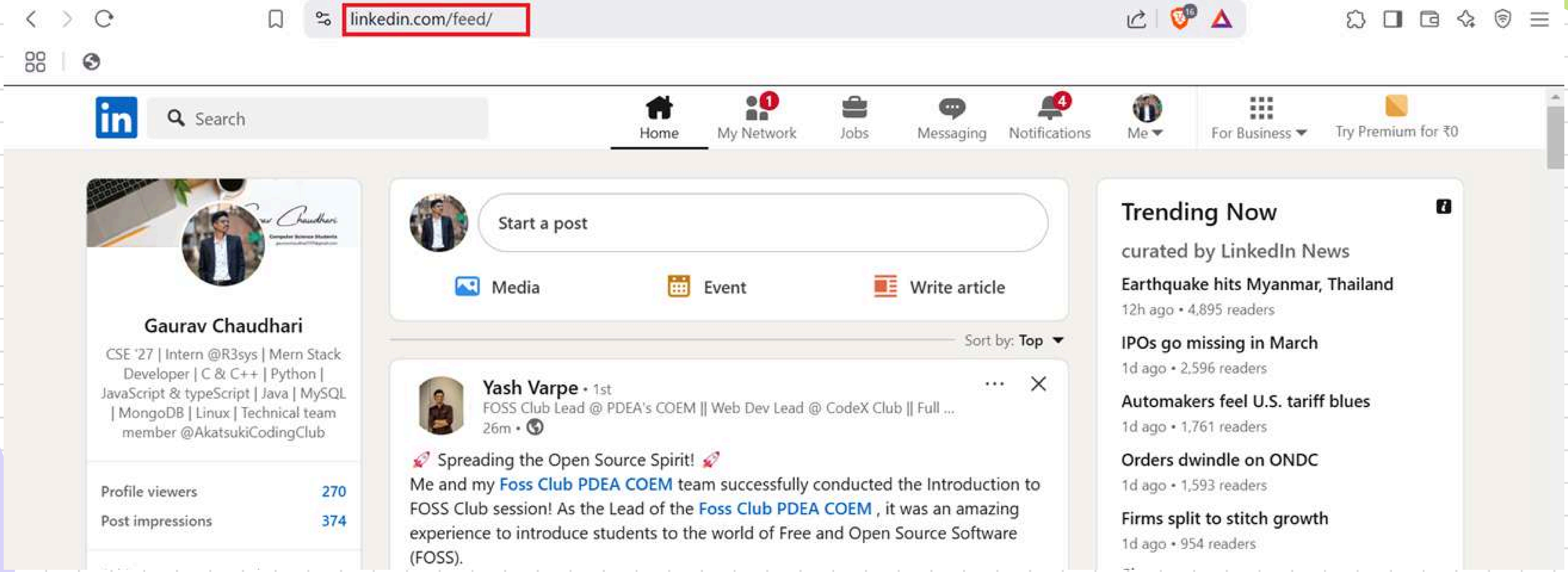
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Me
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Post impressions 374
Start a post
Media
Event
Write article
Sort by: Top
Yash Varpe • 1st
FOSS Club Lead @ PDEA's COEM || Web Dev Lead @ CodeX Club || Full ...
26m •
Spreading the Open Source Spirit!
Me and my Foss Club PDEA COEM team successfully conducted the Introduction to FOSS Club session! As the Lead of the Foss Club PDEA COEM , it was an amazing experience to introduce students to the world of Free and Open Source Software (FOSS).
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REACT-ROUTER-DOM

npm

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react-router-dom TS

7.4.1 • Public • Published 4 days ago

Readme | Code Beta | 1 Dependency | 23,545 Dependents | 632 Versions

This package simply re-exports everything from `react-router` to smooth the upgrade path for v6 applications. Once upgraded you can change all of your imports and remove it from your dependencies:

```
-import { Routes } from "react-router-dom"
+import { Routes } from "react-router"
```

Keywords

react router route routing history link

Install

```
> npm i react-router-dom
```

Repository

github.com/remix-run/react-router

Homepage

github.com/remix-run/react-router#re...

Weekly Downloads

11,952,704

ROUTER CREATION?

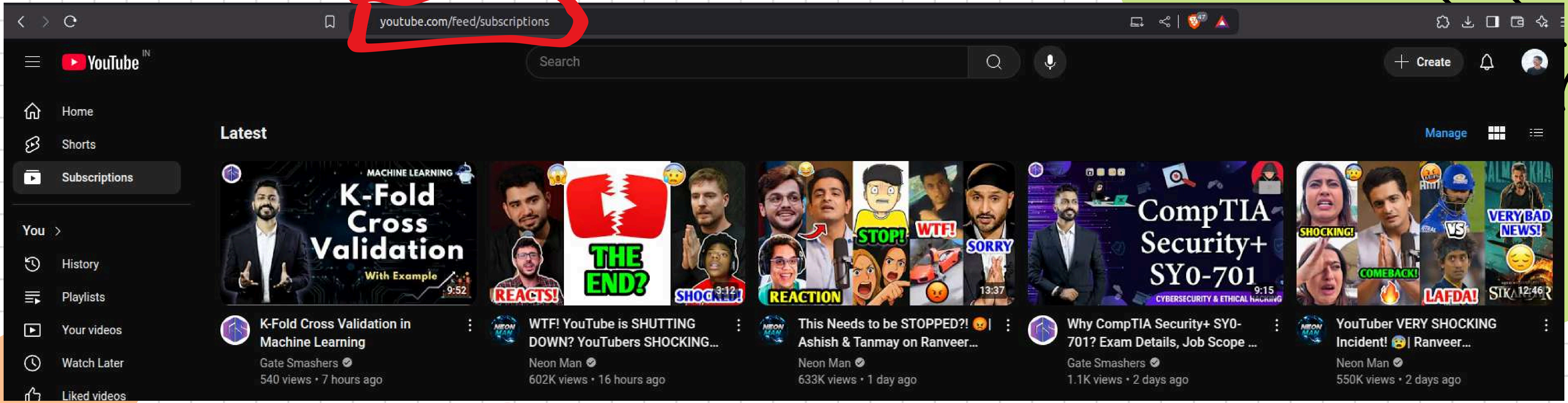
Browser Router

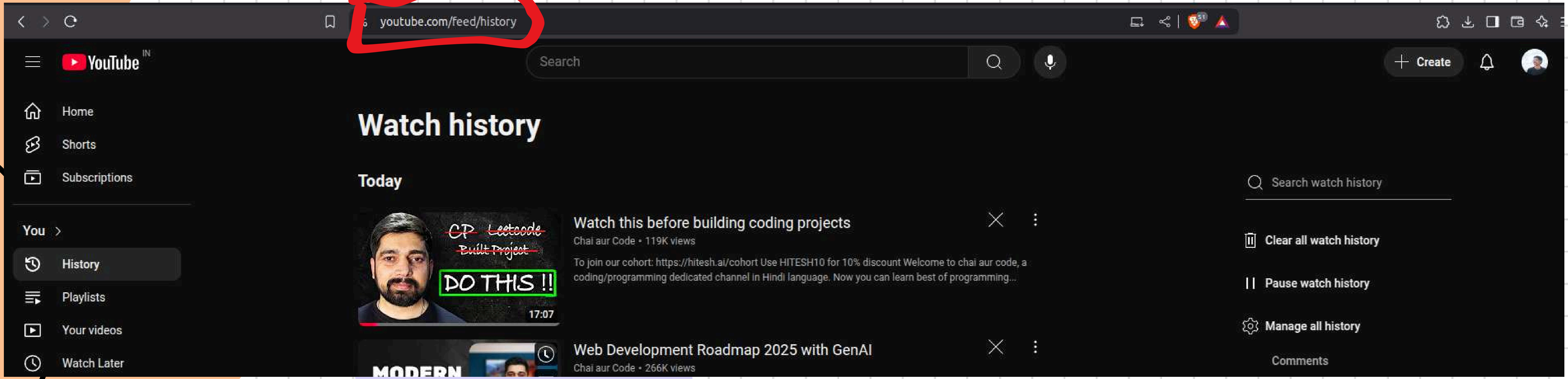
Create Browser Router

ROUTING USING BROWSER ROUTER

```
function App() {

  return (
    <BrowserRouter>
      <Routes>
        <Route path="/dashboard" element={<Dashboard />} />
        <Route path="/" element={<Landing />} />
      </Routes>
    </BrowserRouter>
  )
}
```

NESTED ROUTING



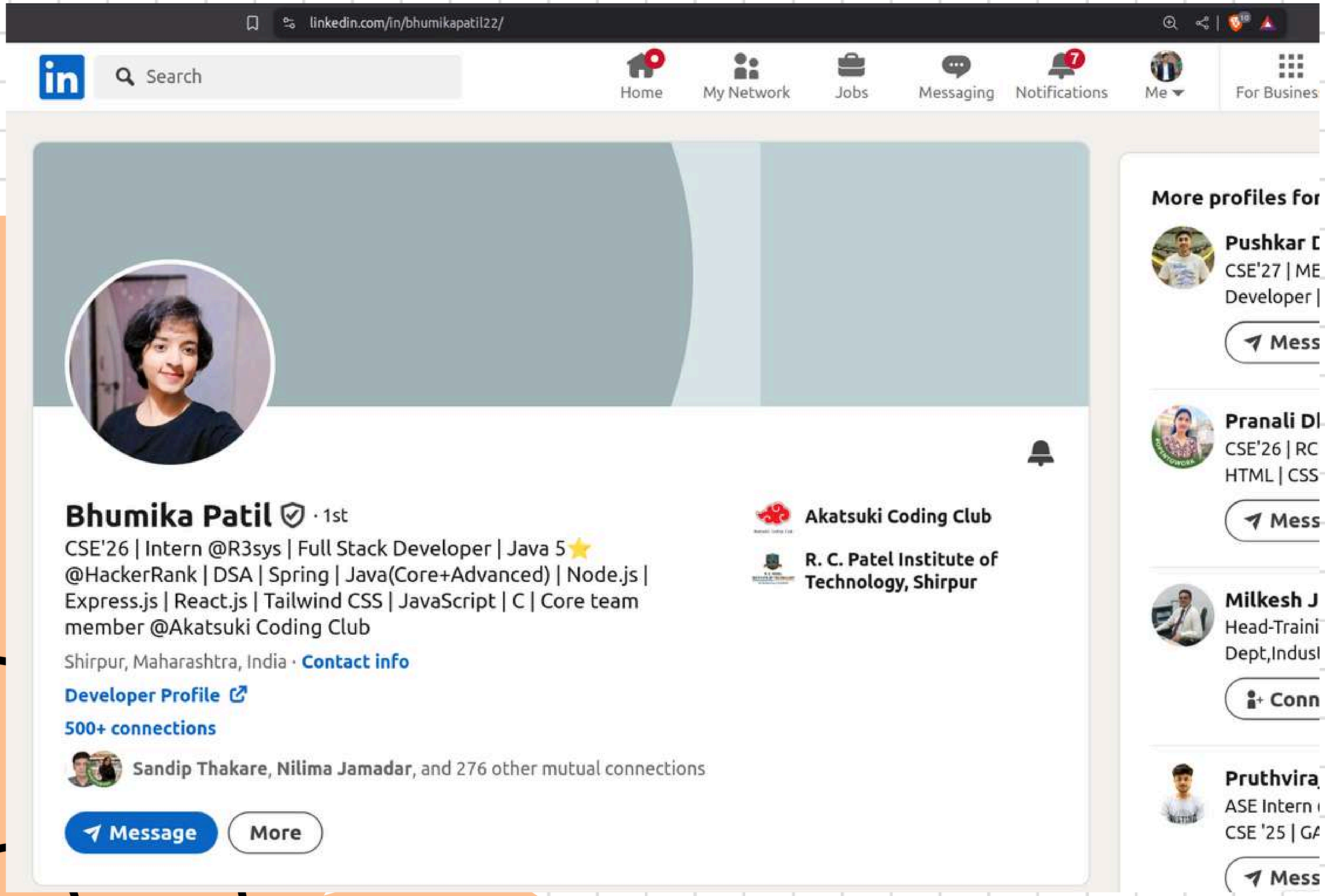


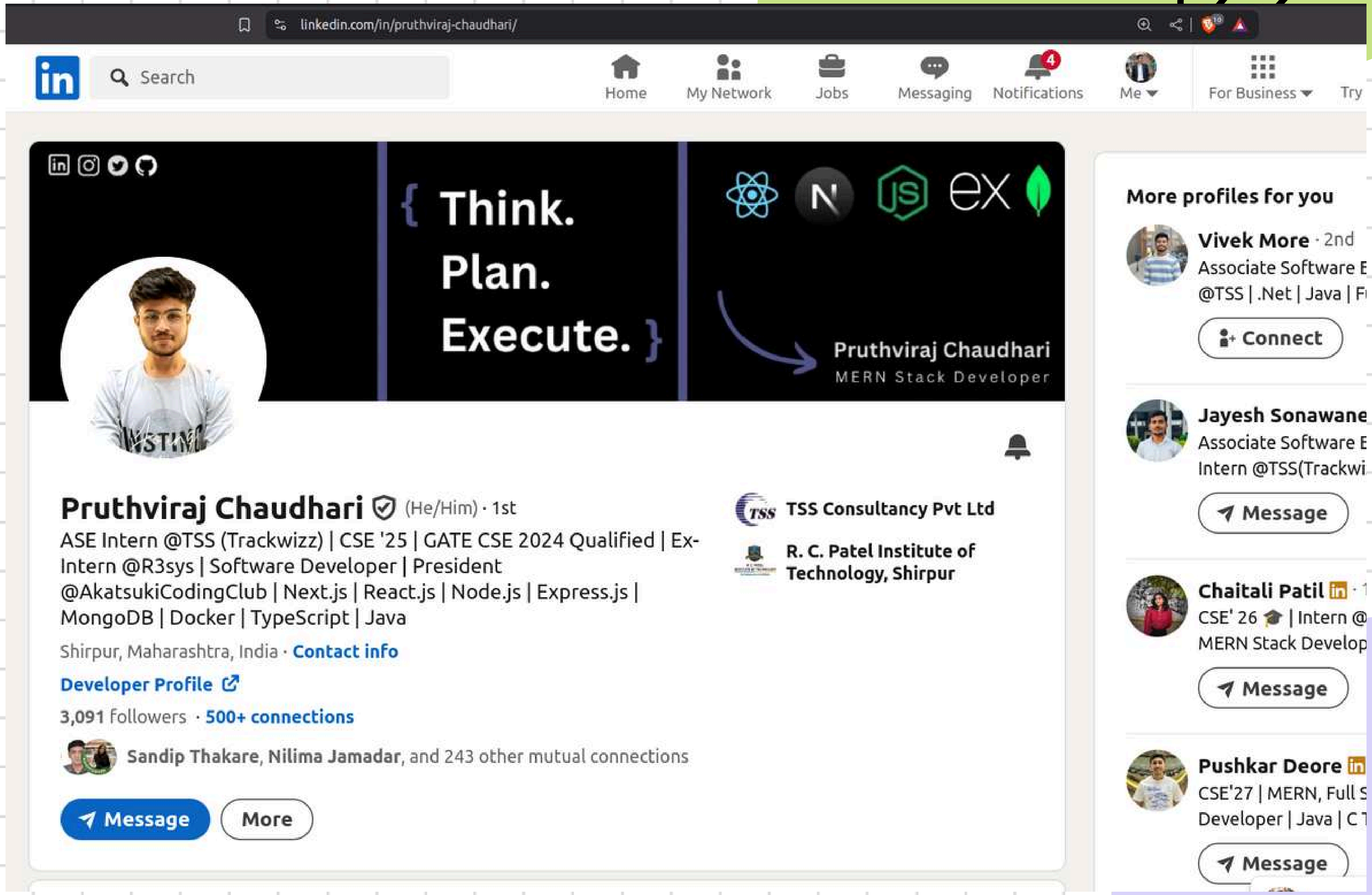
HOW TO CREATE NESTED ROUTES?

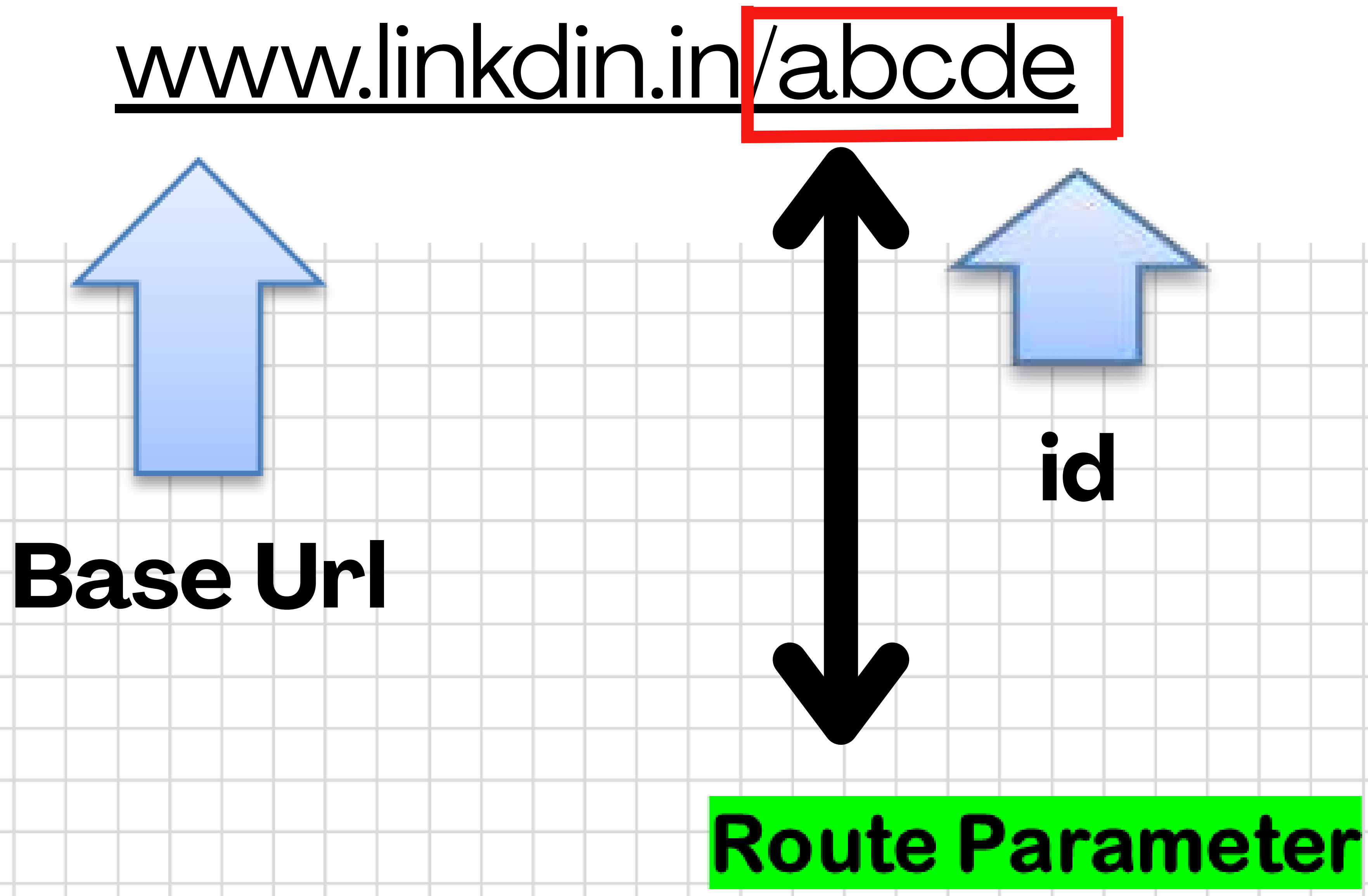
```
const App = () => {
    return (
        <Router>
            <Routes>
                <Route path="/" element={<YouTube />}>
                    <Route path="subscriptions" element={<Subscriptions />} />
                    <Route path="history" element={<History />} />
                </Route>
                <Route path="*" element={<NoMatch />} />
            </Routes>
        </Router>
    );
};

export default App;
```

```
import { BrowserRouter as Router, Routes, Route, Link, Outlet } from 'react-router-dom';

const YouTube = () => {
    return (
        <>
            <h1>YouTube</h1>
            <nav>
                <Link to="subscriptions">Subscriptions</Link>
                <Link to="history">History</Link>
            </nav>
            <Outlet />
        </>
    );
};
```

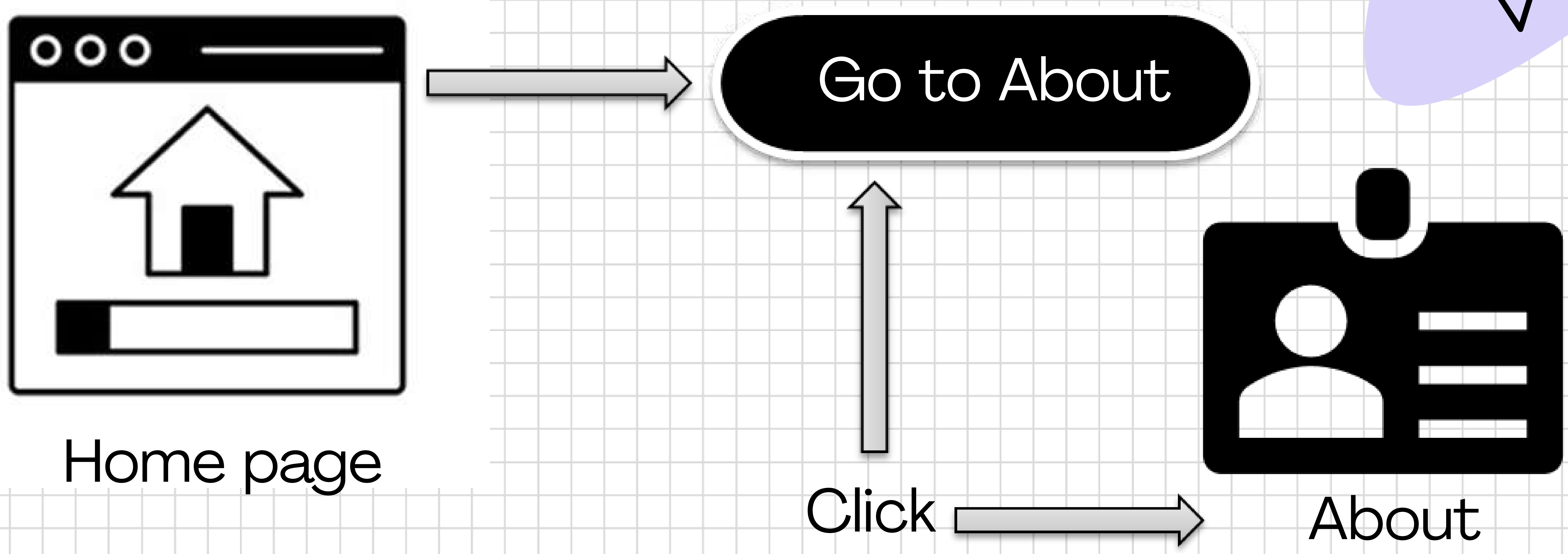
DYNAMIC ROUTING





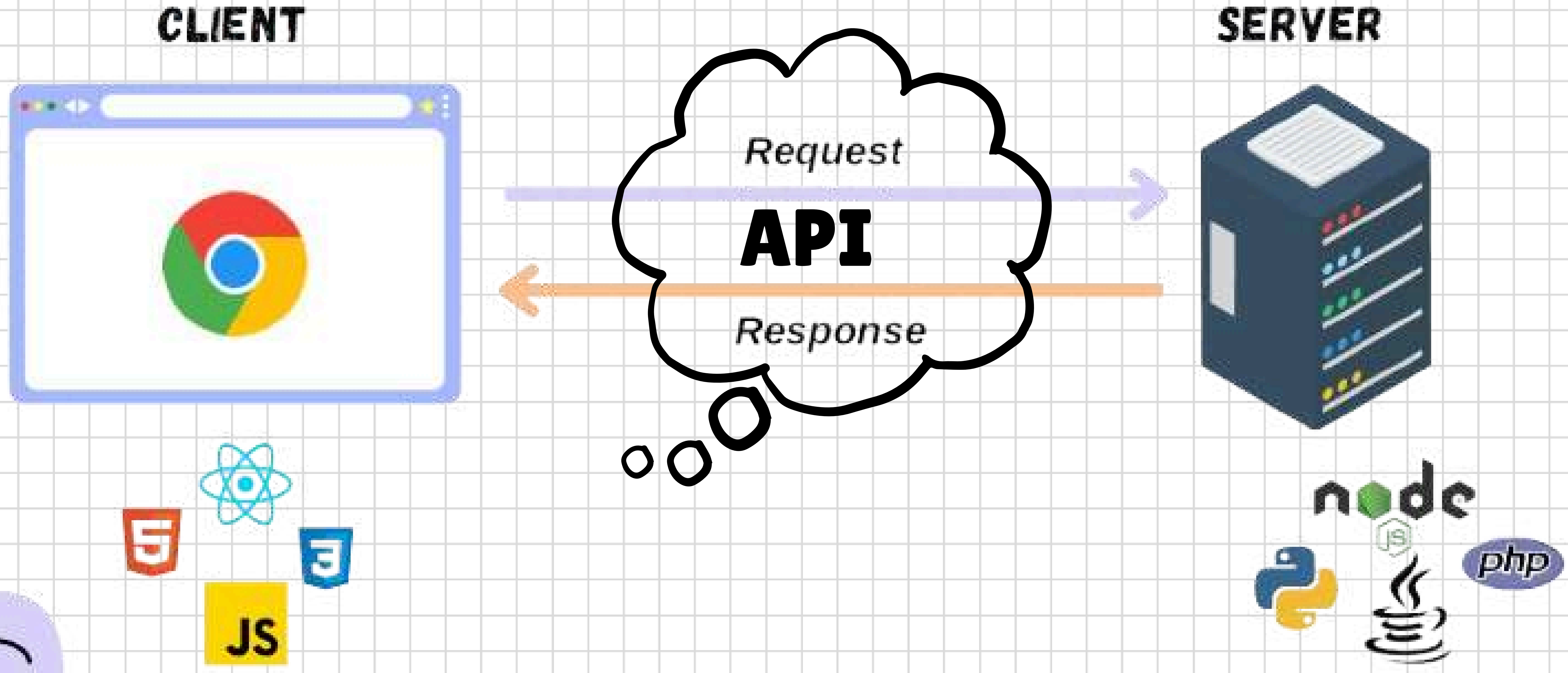

www.linkdin.in/abcde
Base Url
id
Route Parameter

useNavigate



WHAT,HOW & WHY DATA FETCHING????

MECHANISM



USEEFFECT HOOK

```
useEffect(() => {
  /* Code to run when the component
  mounts or when
  specified dependencies change */
}, [dependencies]);
```

USEEFFECT HOOK

AXIOS

```
import axios from 'axios';

axios.get('https://api.example.com/data', {
  headers: {
    'Content-Type': 'application/json'
  }
})
  .then(response => console.log(response.data))
  .catch(error => console.error('Error:', error));
```

FETCH

VS

AXIOS

```javascript
// api calling
 const fetchData = ()=>{
   fetch('https://randomuser.me/api/').then( (data)=>{
     //console.log(data);
     return data.json()
   }).then( (jsonData)=>{
     console.log(jsonData);
   } )
   .catch( (error)=>{
     console.log("error ala bho..");
   })
 }
 fetchData()
```

```javascript
//axios
async function fetching() {
    try {
        let responce = await axios.get('https://randomuser.me/api/')
        console.log(responce.data);
    } catch (error) {
        console.log("Can not fetch data",error);
    }
}
fetching();
```

- Automatic JSON parsing
- Simplified syntax
- Robust error handling

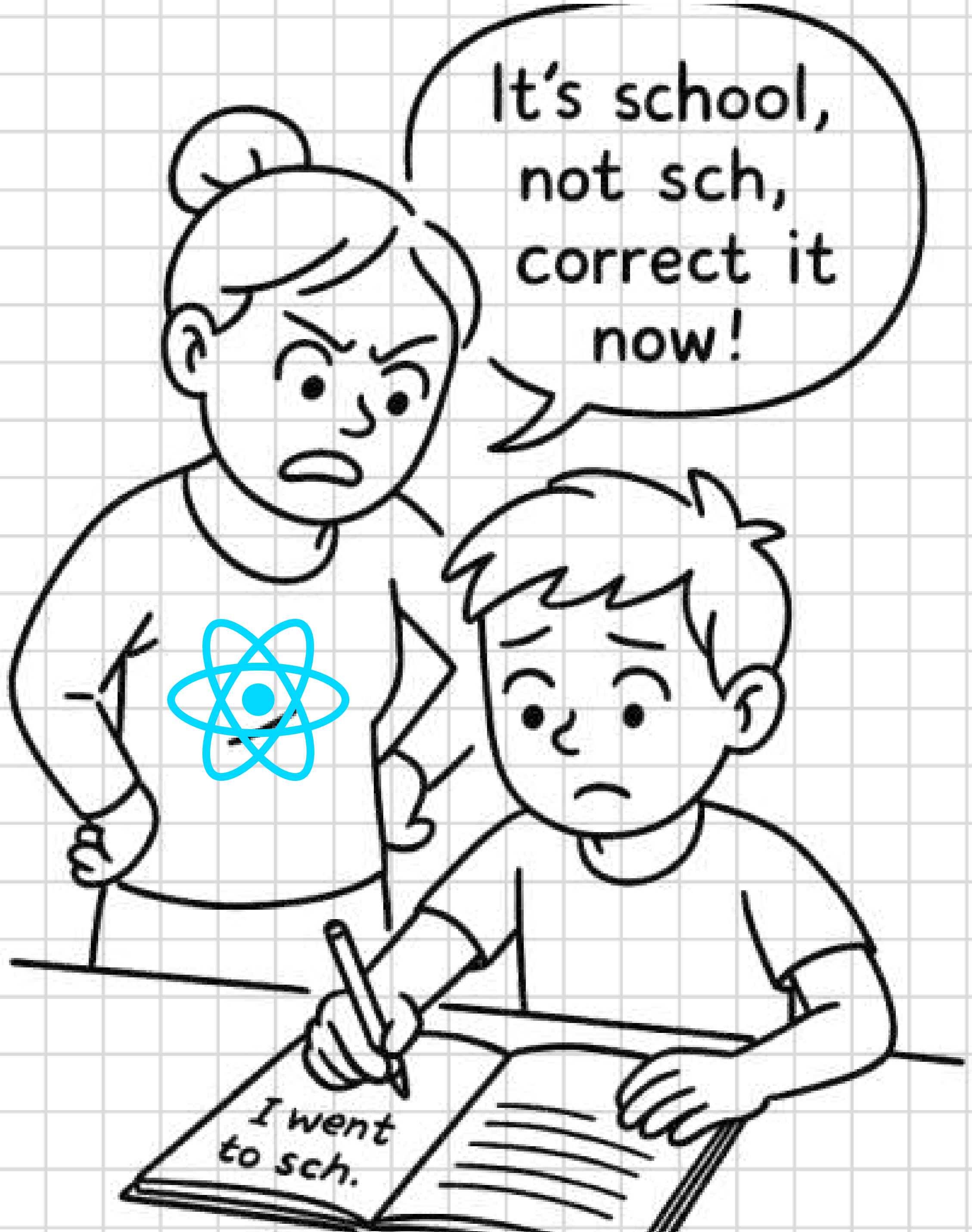
- Requiring installation
- Large bundle size

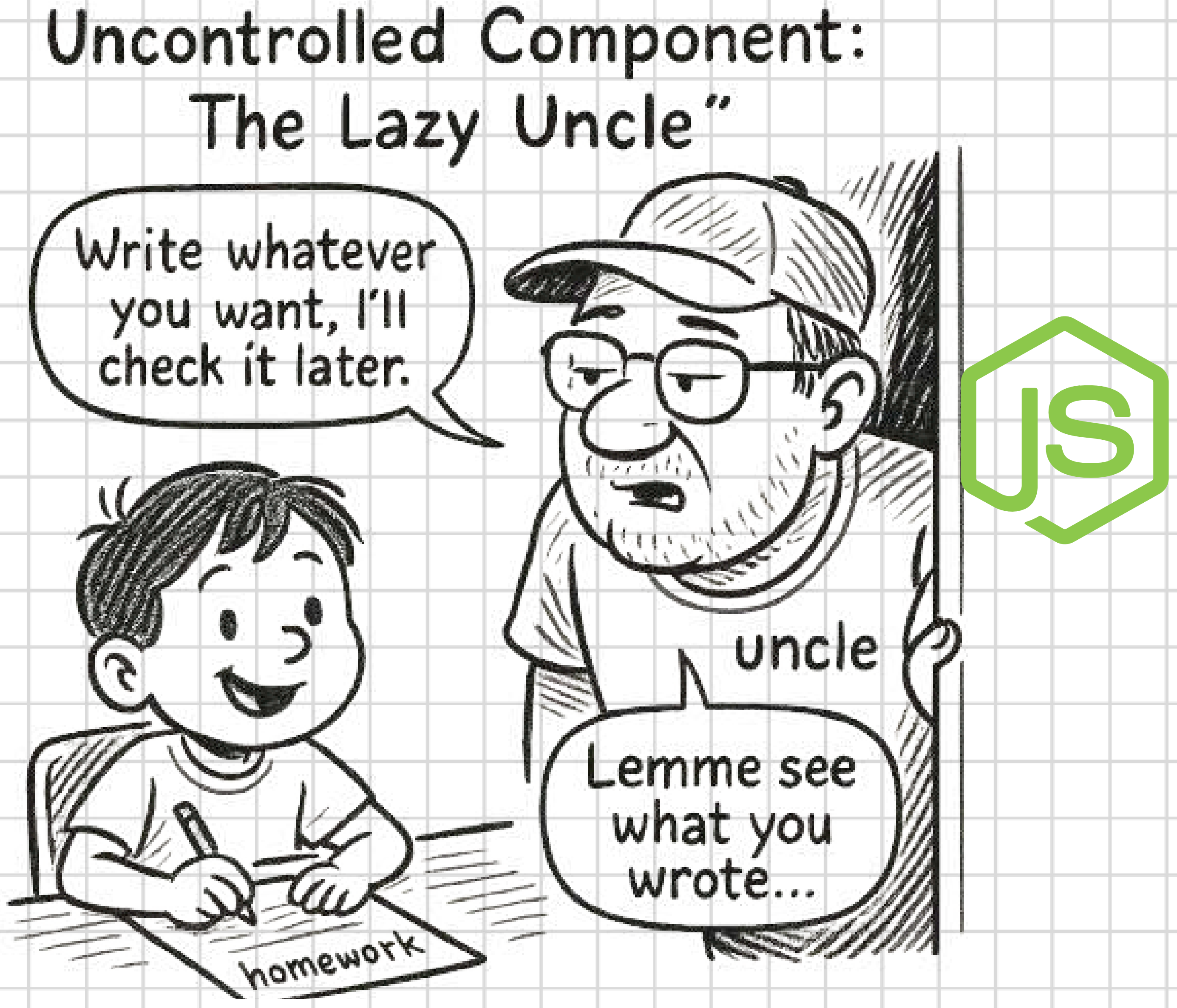
WHAT IS FORM ?? &
HOW WE CAN CREATE IT?......

APPROACH

CONTROLLED COMPONENT

UNCONTROLLED COMPONENT





FORMS HANDLING

State Management

Handles the real time input updates

Event Handlers

Handles the Form Submission

State Updates

Use to stop Default Reload

HANDLING MULTIPLE INPUTS

Use a single state object to manage multiple form fields.

```
const [formData, setFormData] = useState({ name: '', email: ''

const handleChange = (e) => {
  setFormData({ ...formData, [e.target.name]: e.target.value });
};
```